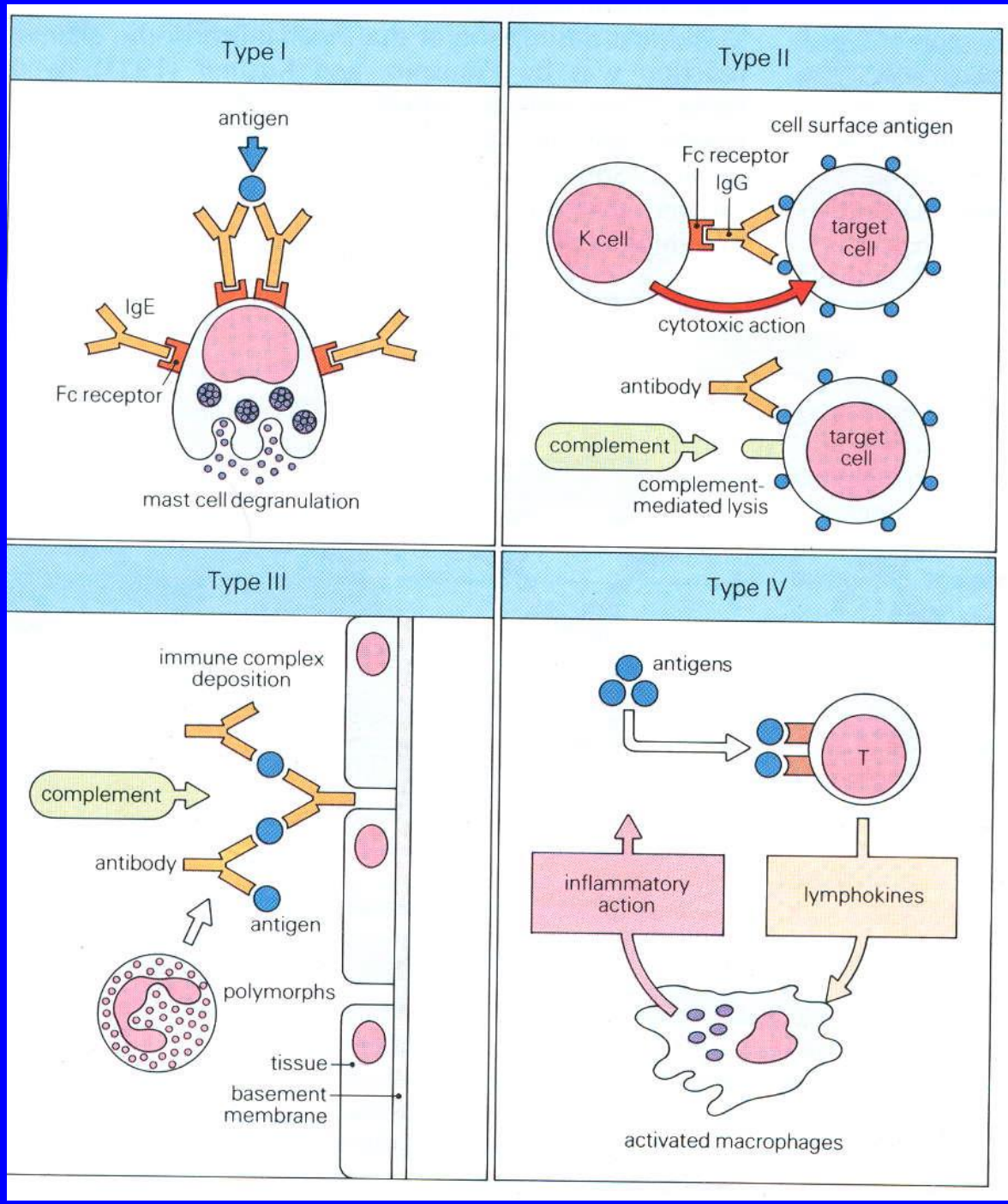
Hipersensibilidades

Gell e Coombs

Hipersensibilidade

Dano tecidual imunologicamente mediado

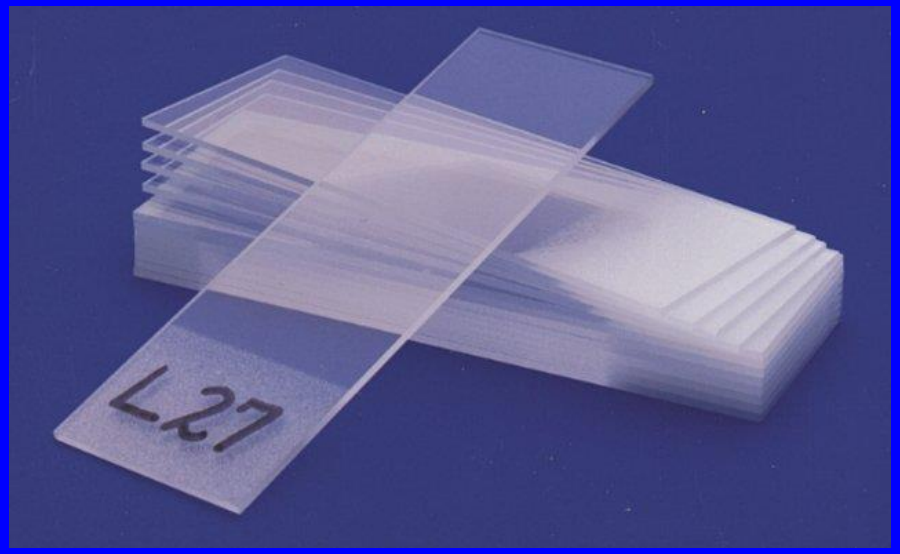


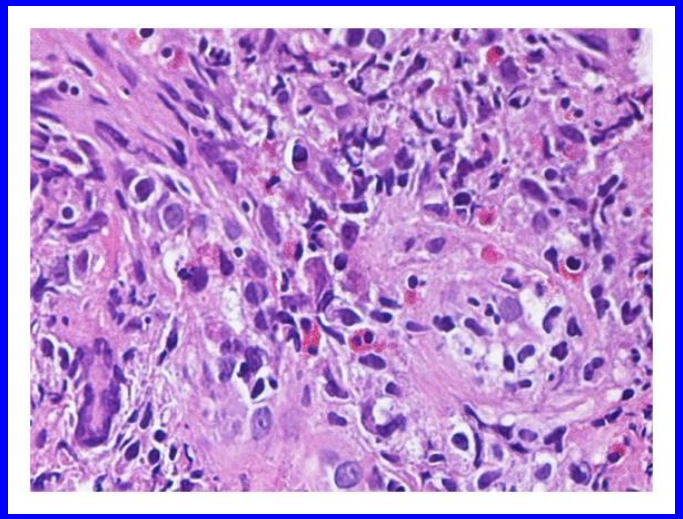
Hipersensibilidade tipo II

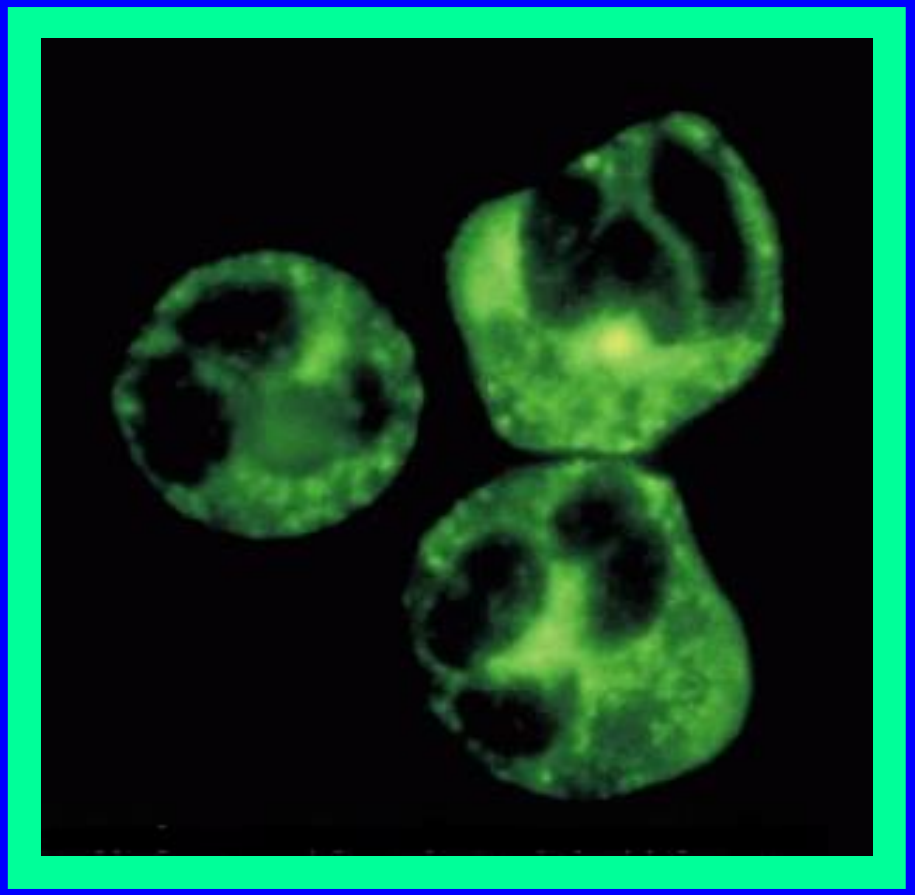
hipersensibilidade por Anticorpos IgM ou IgG dirigidos frente a estruturas celulares ou da matriz extracelular....

.....que afetam a fisiologia celular ou tissular

Estes anticorpos podem ser revelados (normalmente por imunofluorescência) incubando o soro do paciente com cortes do tecido que é o suposto alvo dos anticorpos (tecido obtido na autopsia de outro individuo)

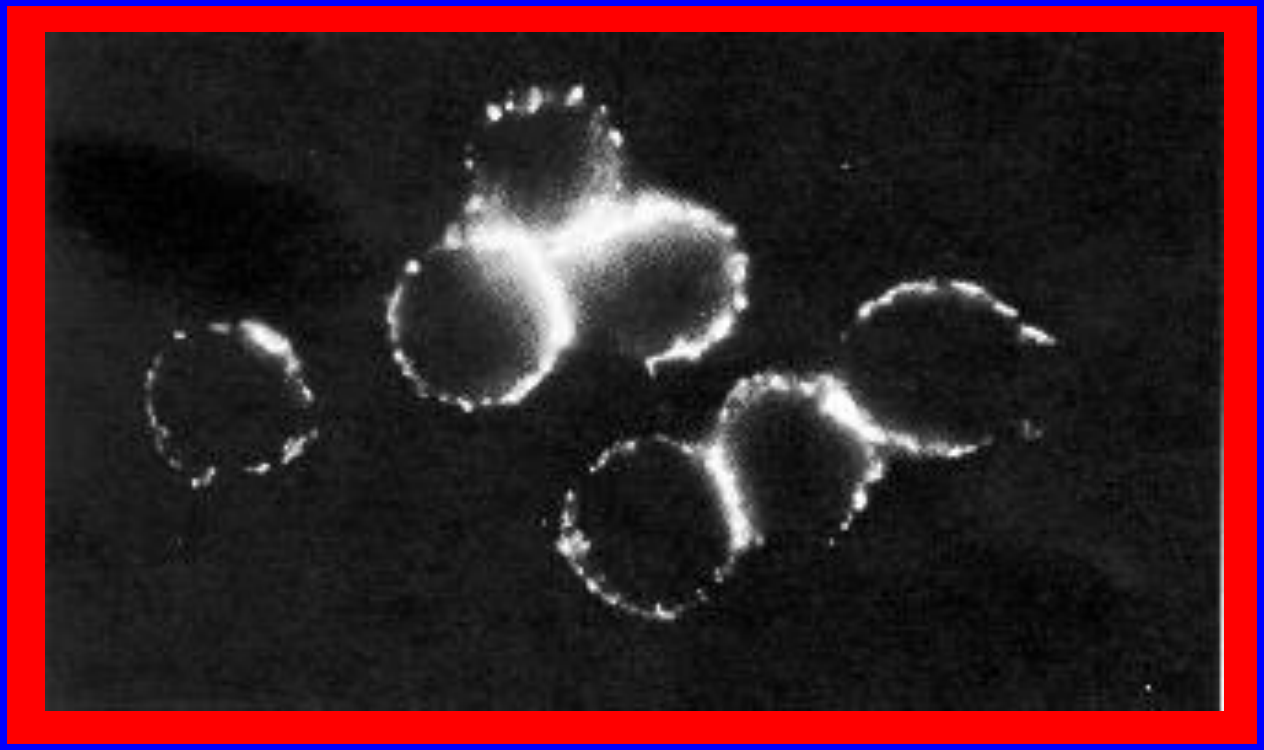




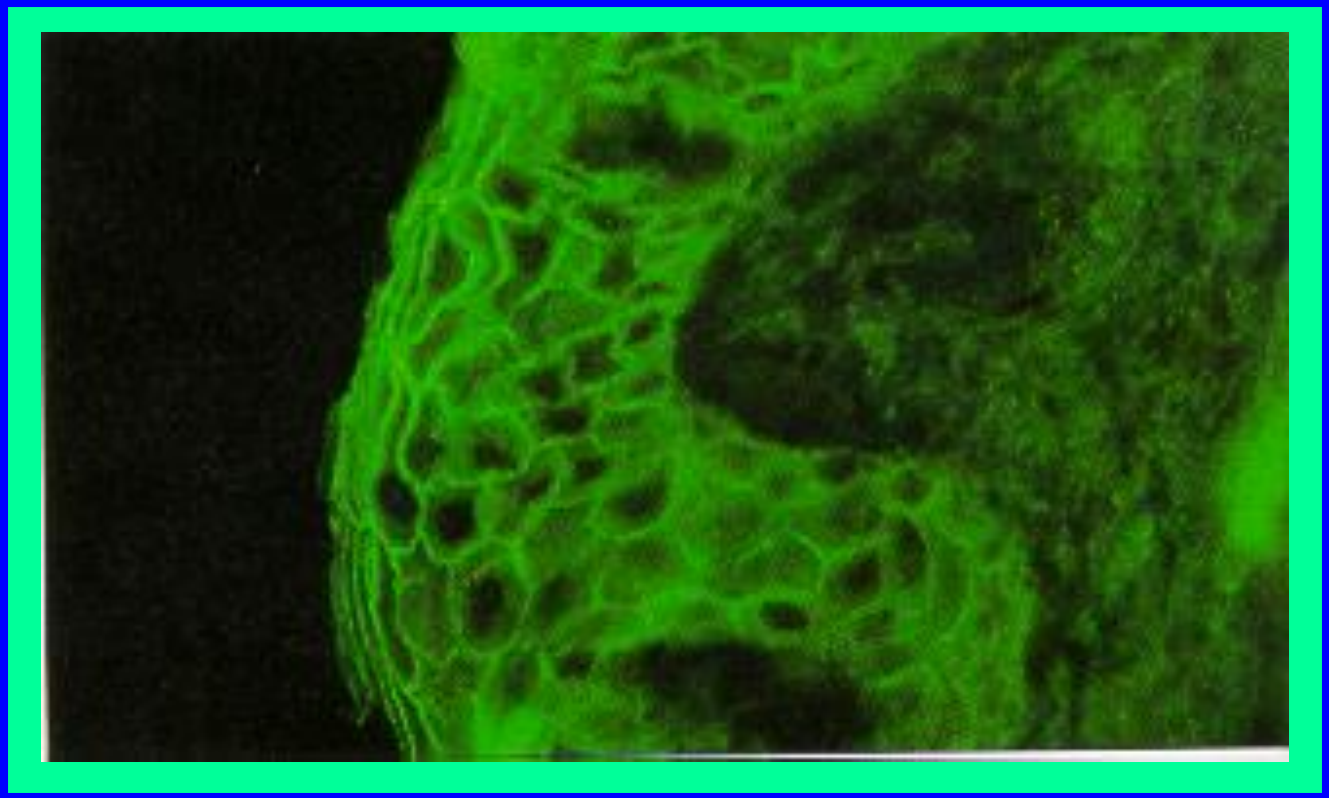


Hipersensibilidade tipo II. **Anticorpos anti-citoplasma do neutrófilo** em paciente com Granulomatose de Wegener

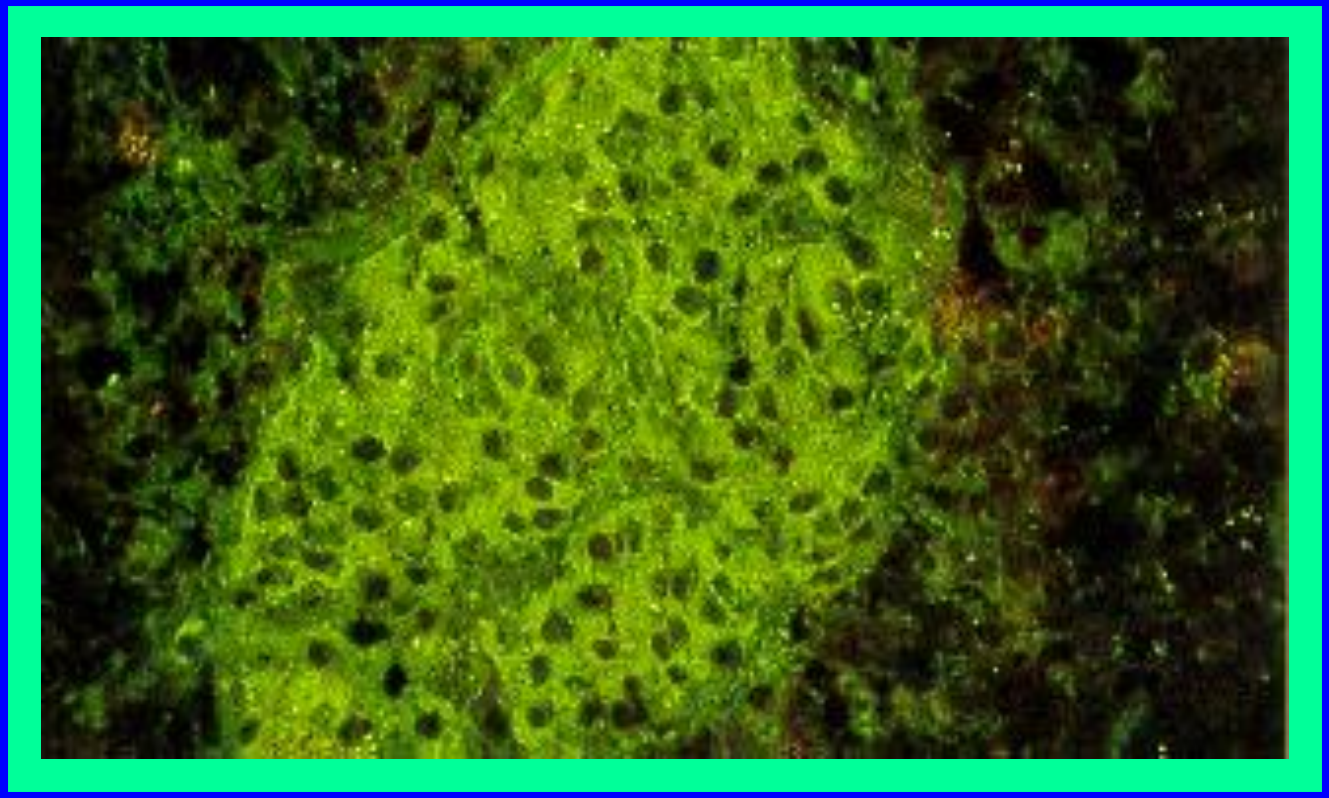
Hipersensibilidade tipo II
Anticorpos anti-neutrófilo (tipo II) em paciente com lupus (tipo III)



Hipersensibilidade tipo II. Anticorpos anti-desmossomos da pele em um paciente com pénfigo



Anticorpos anti-ilhotas do pâncreas em um paciente com diabetes mellitus



- Doença hemolítica do RN
- Reações transfusionais
- Anemia hemolítica autoimune
- Púrpura trombocitopénica autoimune
- Pénfigo
- Síndrome de Goodpasture
- Miastenia gravis
- Febre reumática
- Tiroidite de Hashimoto
- Doença de Graves
- Diabete insulino - resistente
- Anemia perniciosa
-Doença de Adisson
- Miocardite autoimune
-Infertilidade espontânea

DOENÇA	REATIVIDADE DOS ANTICORPOS
-Doença hemolítica do RN	RH+
- Reações transfusionais	A, B
- Anemia hemolítica autoimune	Hemácias
- Púrpura trombocitopénica autoimune	Plaquetas
- Pénfigo	Desmogleina (desmossomos)
- Síndrome de Goodpasture	Membrana basal (rim e pulmão)
- Miastenia gravis	Rec. de acetilcolina- placa motora
- Febre reumática	Streptococus (cruza com ag do coração e articulações)
-Tiroidite de Hashimoto	Ags da tireoide
- Doença de Graves (LATS)	Receptor de TSH
- Diabete insulina – resistente	Insulina*
- Anemia perniciosa	Fator intrínseco
-Doença de Adisson	Suprarrenal*
-Infertilidade espontânea	Espermatozoides*

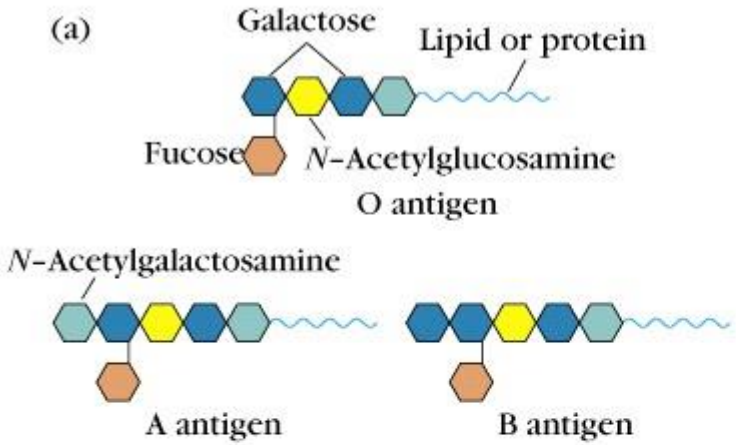
*Em uma percentagem dos casos



Marty Feldman in "Young Frankenstein"

Reações transfusionais

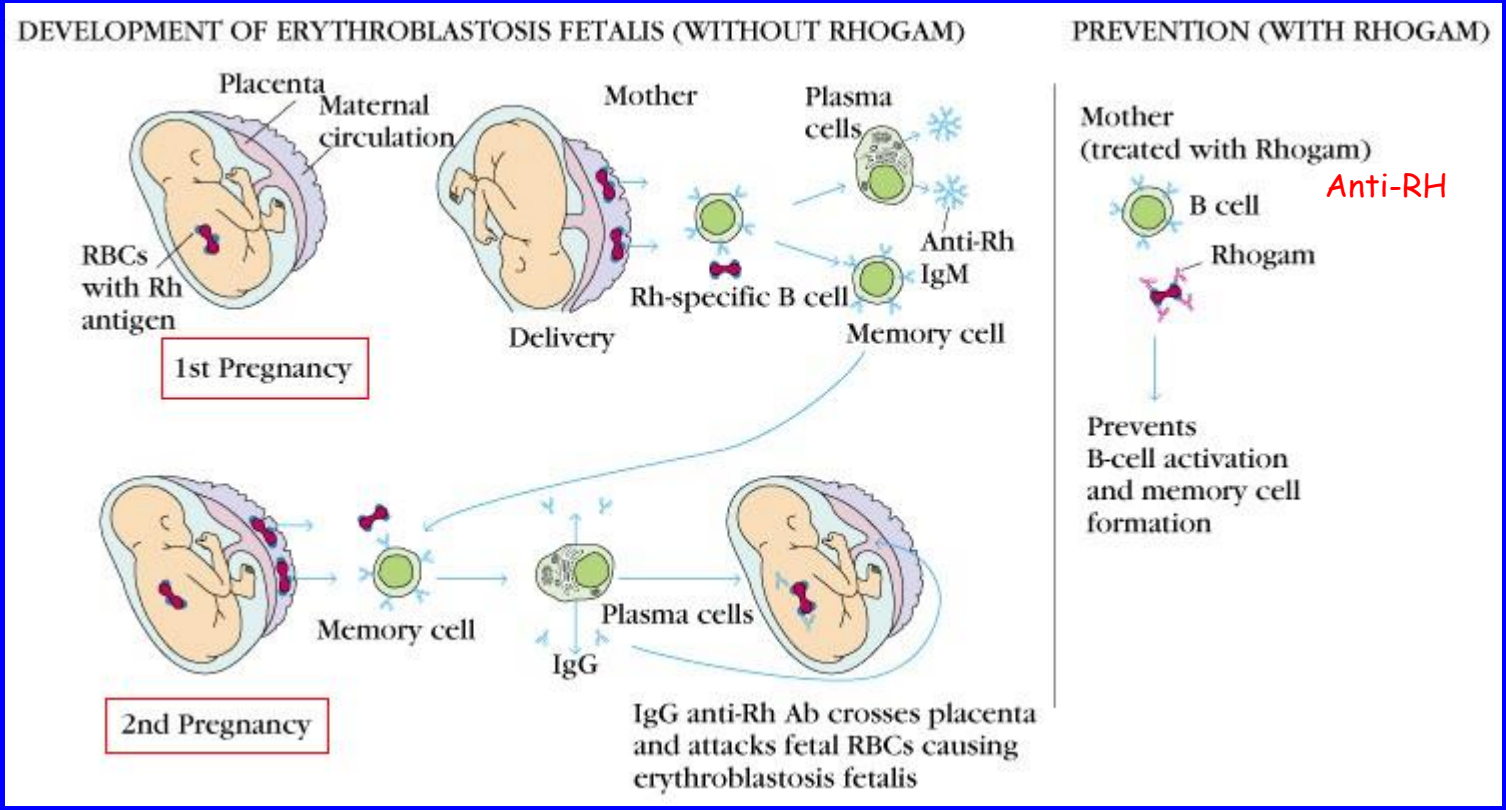
Grupos sanguíneos ABO



(b)

Genotype	Blood-group phenotype	Antigens on erythrocytes (*agglutinins*)	Serum antibodies (*isohemagglutinins*)
AA or AO	A	A	Anti-B
BB or BO	B	B	Anti-A
AB	AB	A and B	None
OO	O	None	Anti-A and anti-B

Grupo Rh: Eritroblastose fetal



Hipersensibilidade tipo III

hipersensibilidade por Anticorpos IgM ou IgG dirigidos contra antígenos solúveis (estranhos ou próprios)

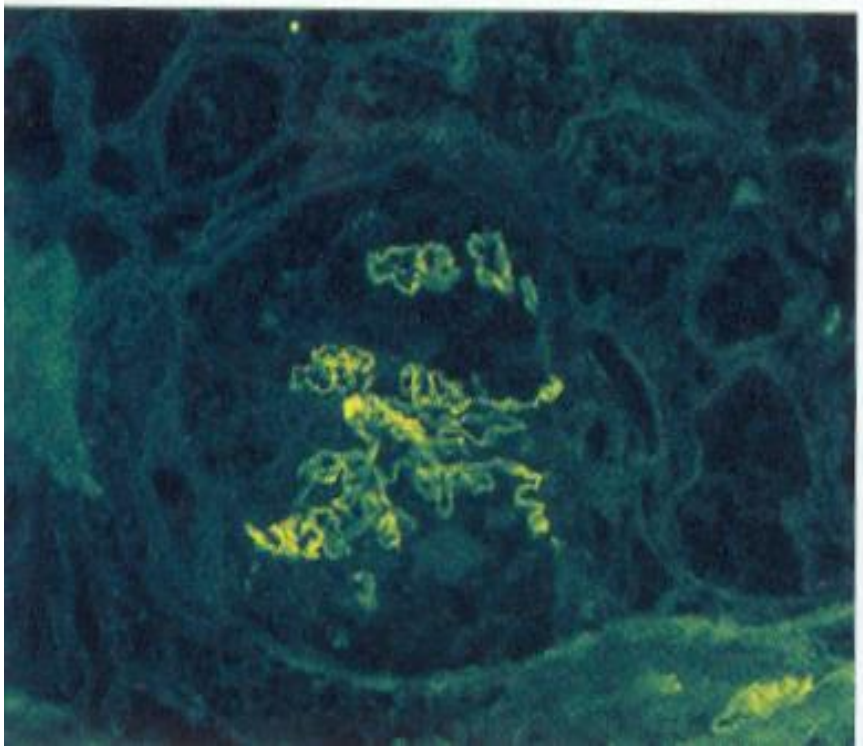
E onde a lesão no paciente ocorre pelo depósito de imunocomplexos (que se encontram em grande quantidade)

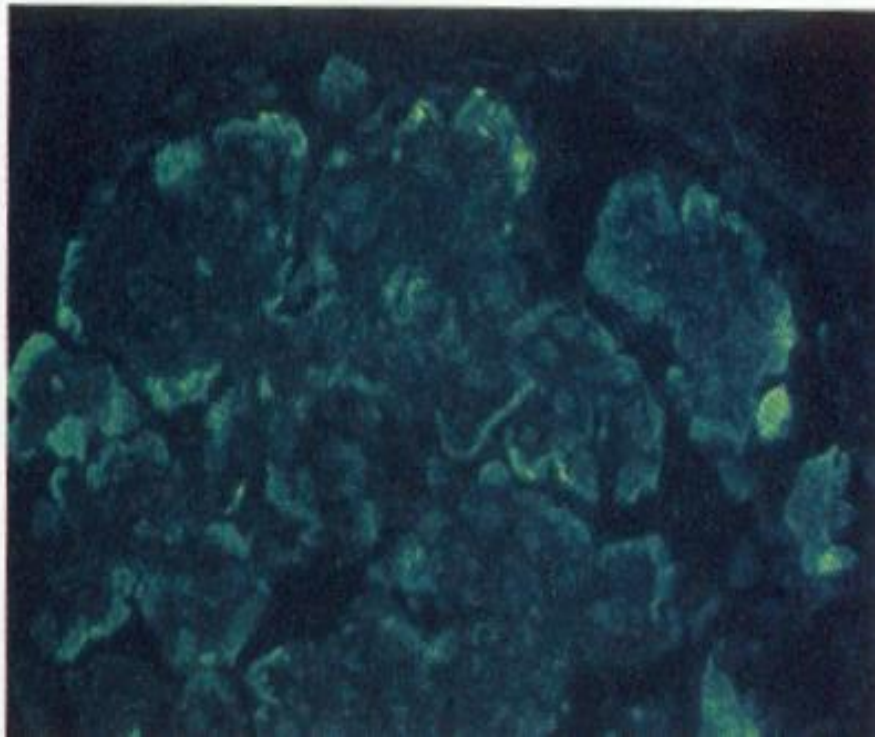
- Doença do soro
- Síndrome nefrótica em Malária
- Glomerulonefrite pós-estreptocócica
- Poliarterite nodosa (hepatite B e outras viroses)
- Elefantíase na filariose

- Lupus eritematoso disseminado
- Artrite reumatoide
-Espondilite anquilosante

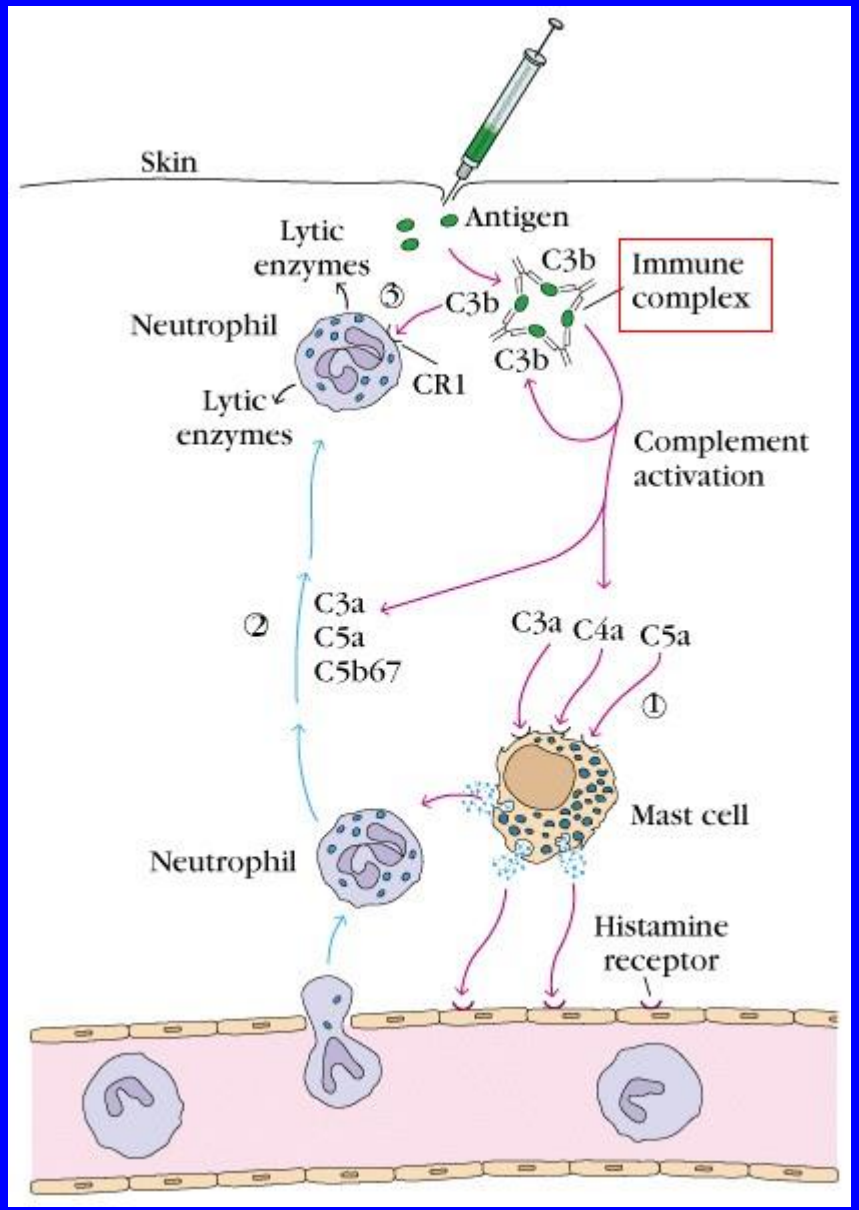
-Reação de Arthus

Imunofluorescência para IgG no glomérulo renal



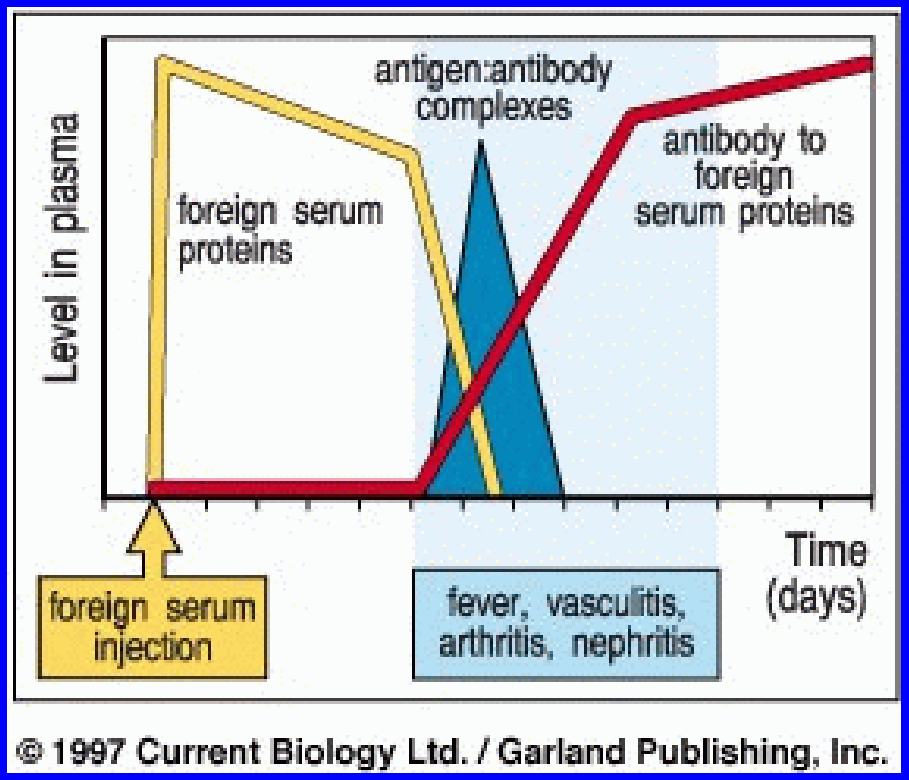


Ação local de imunocomplexos



Ação sistêmica de imunocomplexos:

Deposito na parede dos vasos (vasculite)



Doença do soro (hipersensibilidade tipo III)



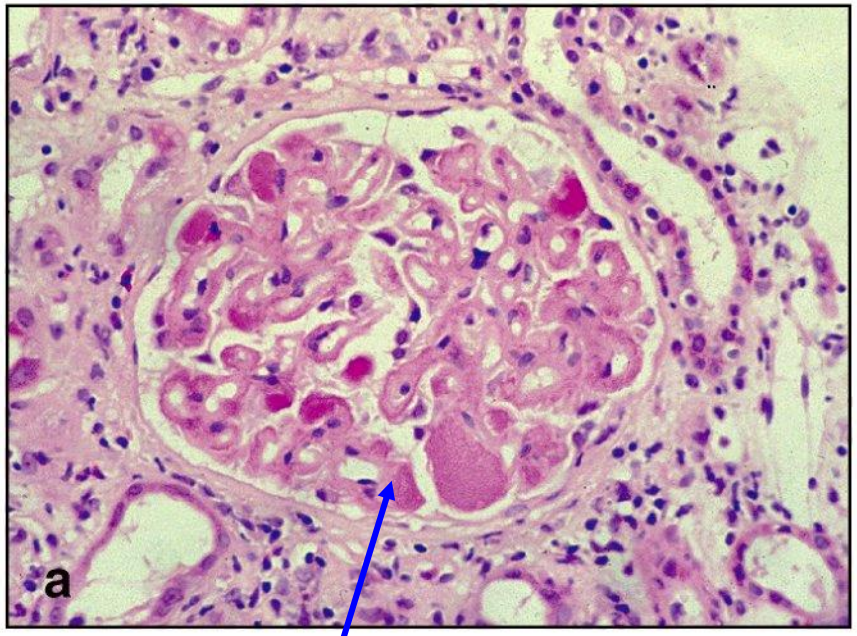
Hemorragia na pele na doença do soro

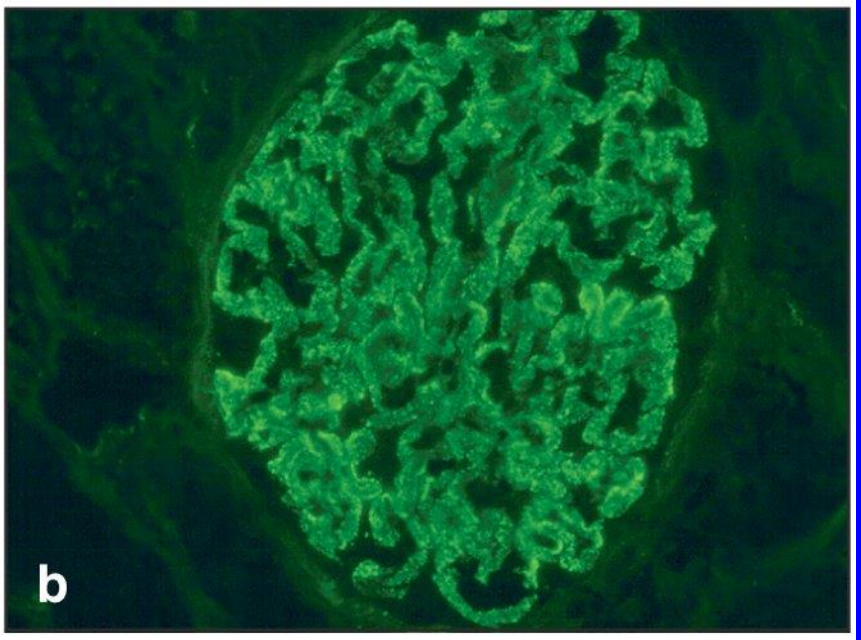


Rash cutâneo na doença do soro

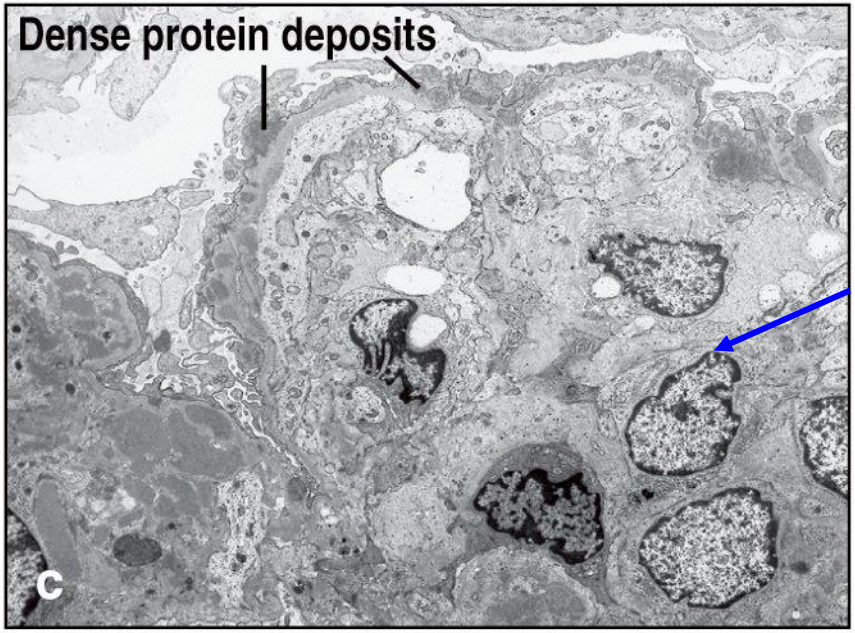


Lupus eritematoso disseminado





Engrossamento da membrana basal



PMN

lupus

Figure 13-33 Immunobiology, 6/e. (© Garland Science 2005)

Lupus eritematoso disseminado

Anticorpos frente a:

- DNA (fita simples ou dupla),
- histonas,
- nucleoproteinas,
- RNA nuclear

Depósito de Imunocomplexos em:

-Vasos (vasculite)

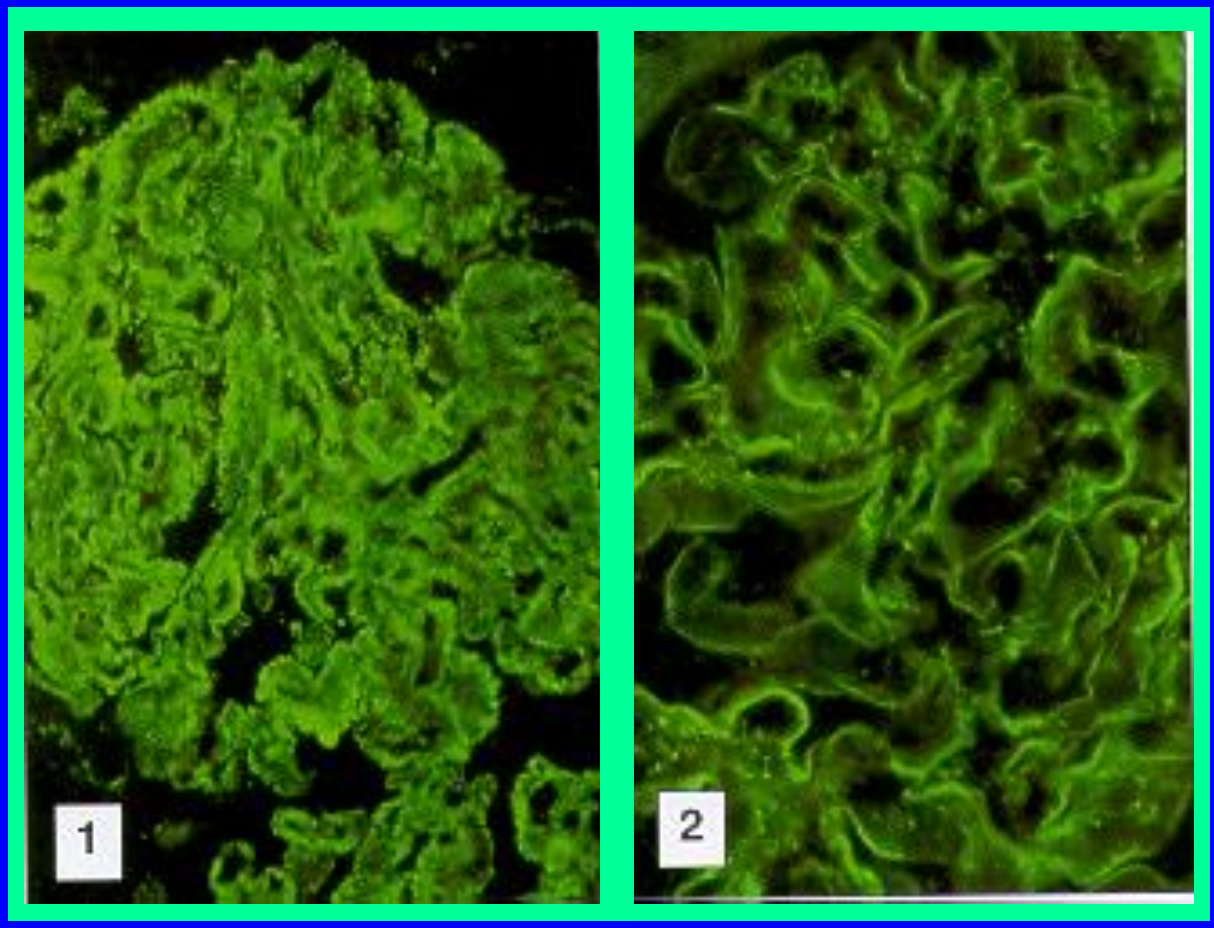
-Glomérulo renal

-Cavidade articular

-Pele

Lupus eritematoso disseminado

Imunofluorescência para IgG no glomérulo renal



Complexos imunes (SLE) Tipo III

Anticorpos Anti-membrana basal (Goodpasture) Tipo II

Hipersensibilidade tipo IV ou hipersensibilidade Tardia

hipersensibilidade mediada por células T (CD4+ ou CD8+)

- dermatite de contato (Ni, Cr, Latex, poison ivy)

- doenças granulomatosas
- erupções cutâneas da varicela, herpes, sarampo

- IDDM
- Esclerose Múltipla
- Artrite reumatoide
- Doença de Crohn
- Cirrose biliar primária

"delayed type hypersensitivity"

Hipersensibilidade de contato (Tipo IV)
Dermatite por contato





Talvez seja mais TH2 ou TH17, do que TH1
Quando vira crônica se denomina eczema

Hipersensibilidade de contato (Tipo IV)



Borracha do latex



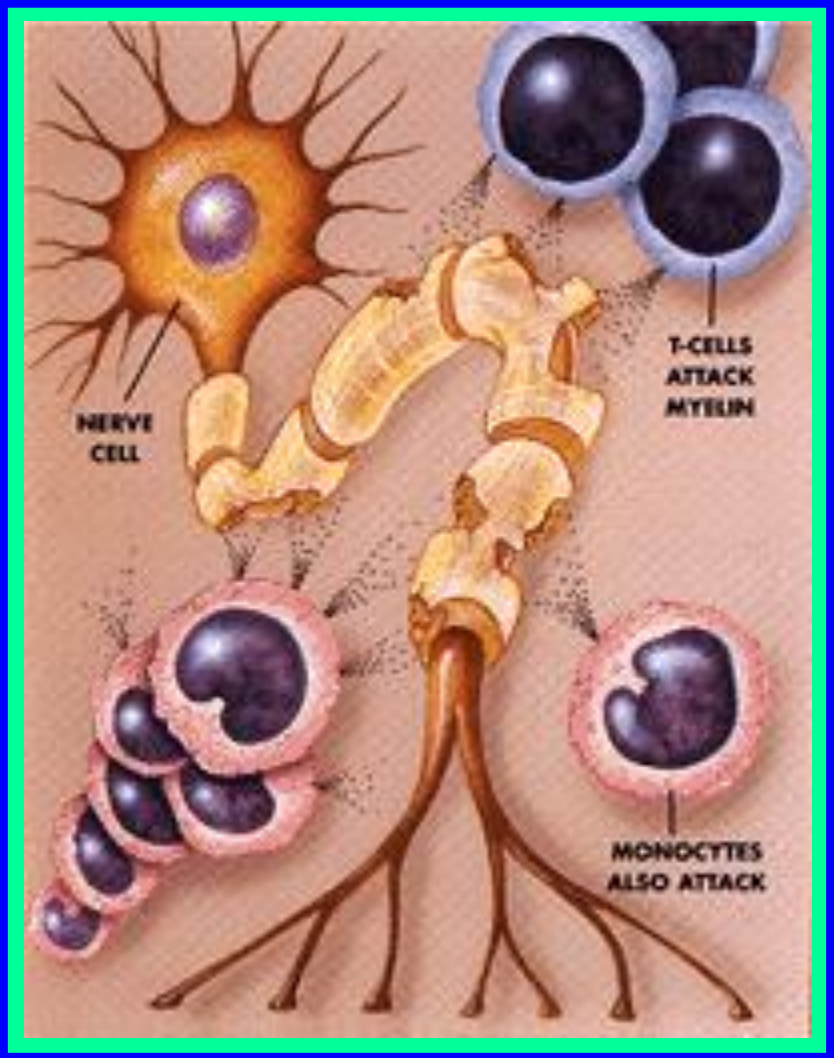
Hipersensibilidade de contato ao níquel (Tipo IV)

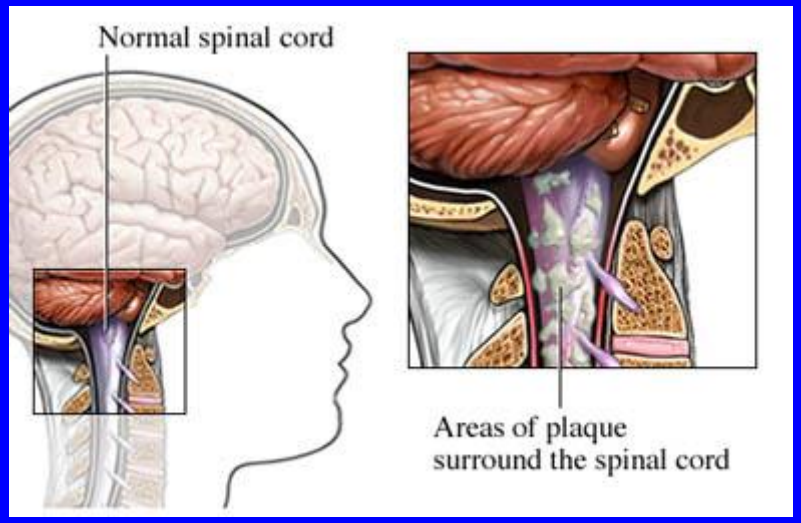




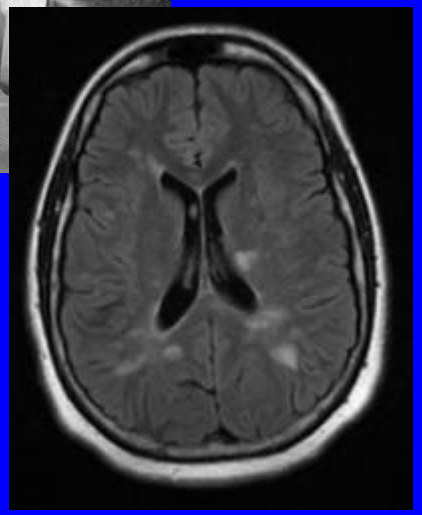
Dermatite por contato induzida por "poison ivy"
Hera venenosa (pentadecacatecol) (reação Tipo IV)

Esclerose múltiple



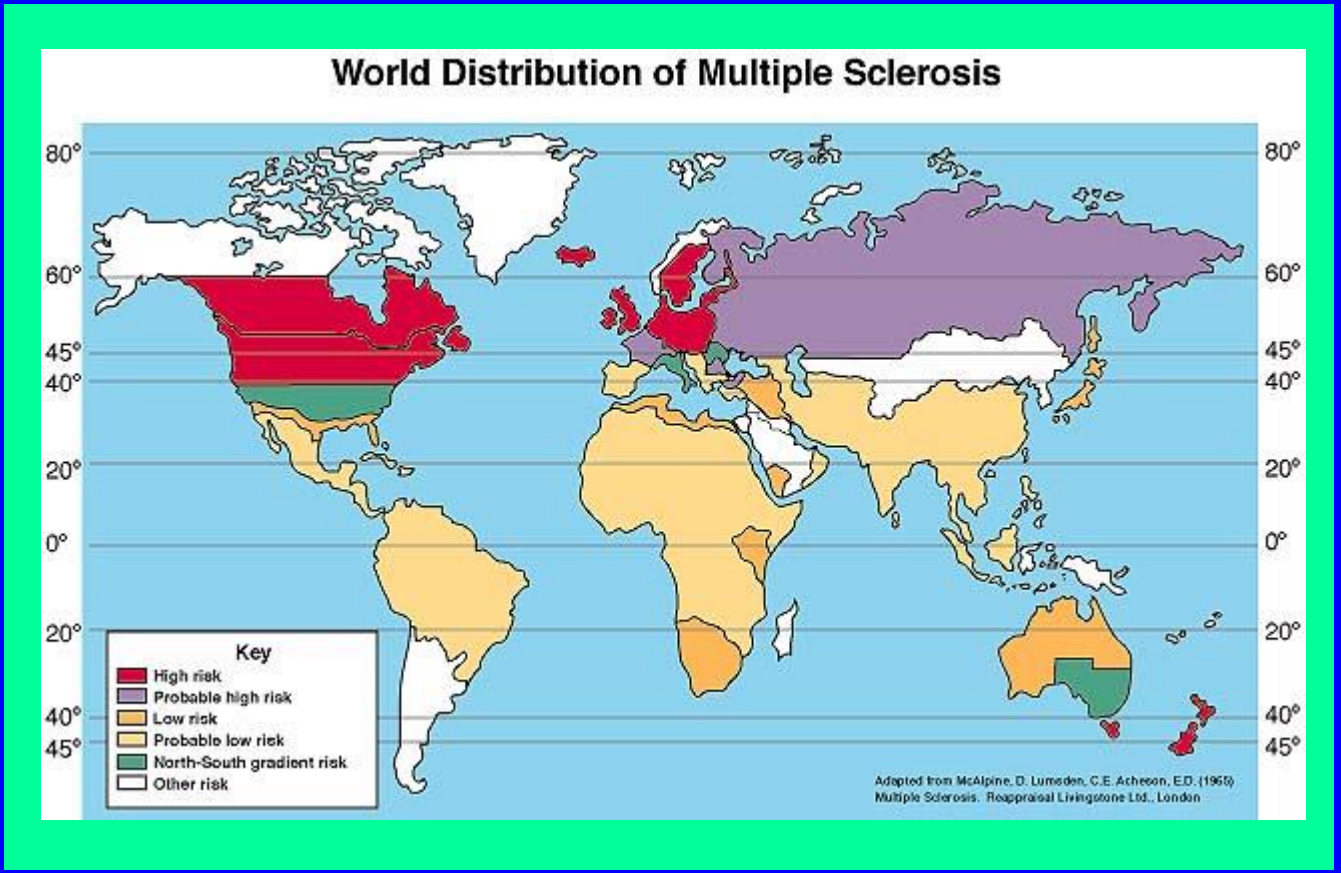






Células T anti-mielina

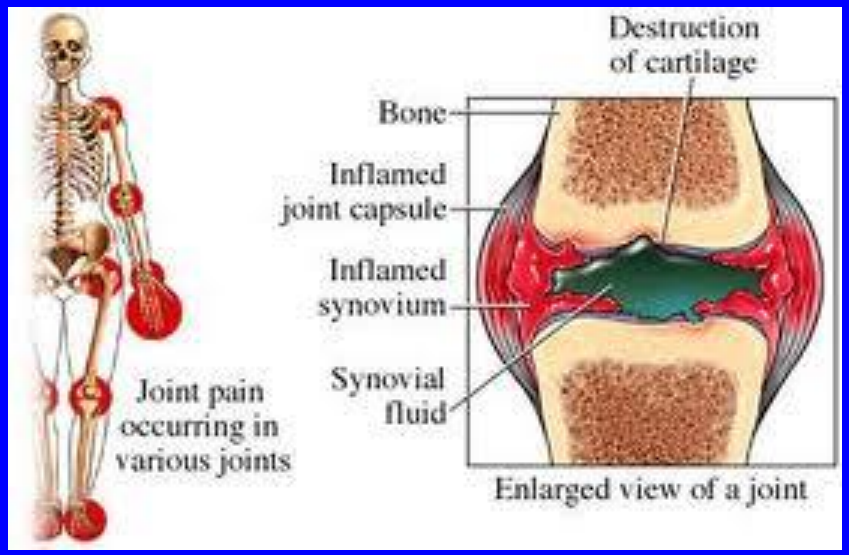
Esclerose múltiple



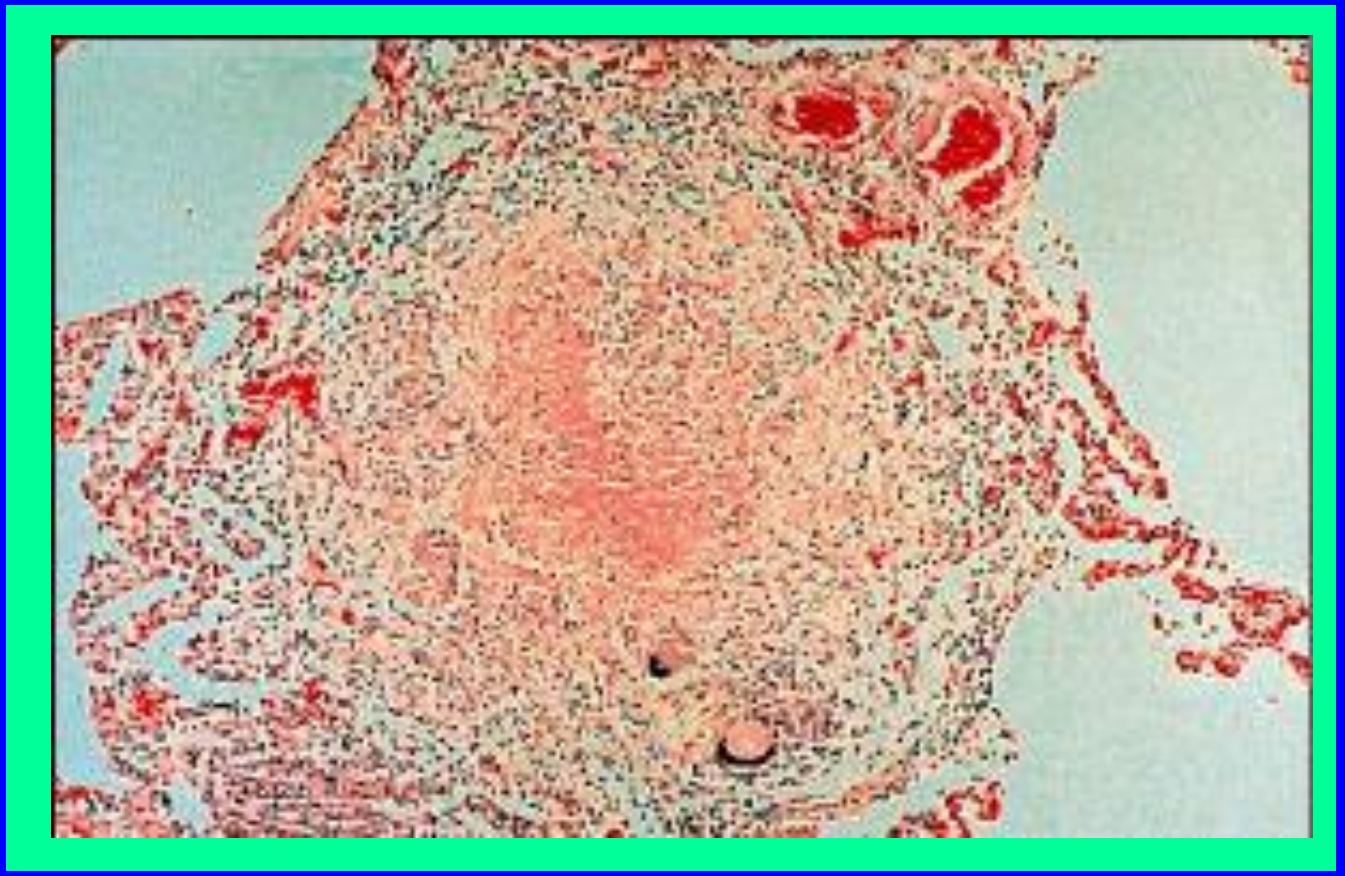
Artrite Reumatoide



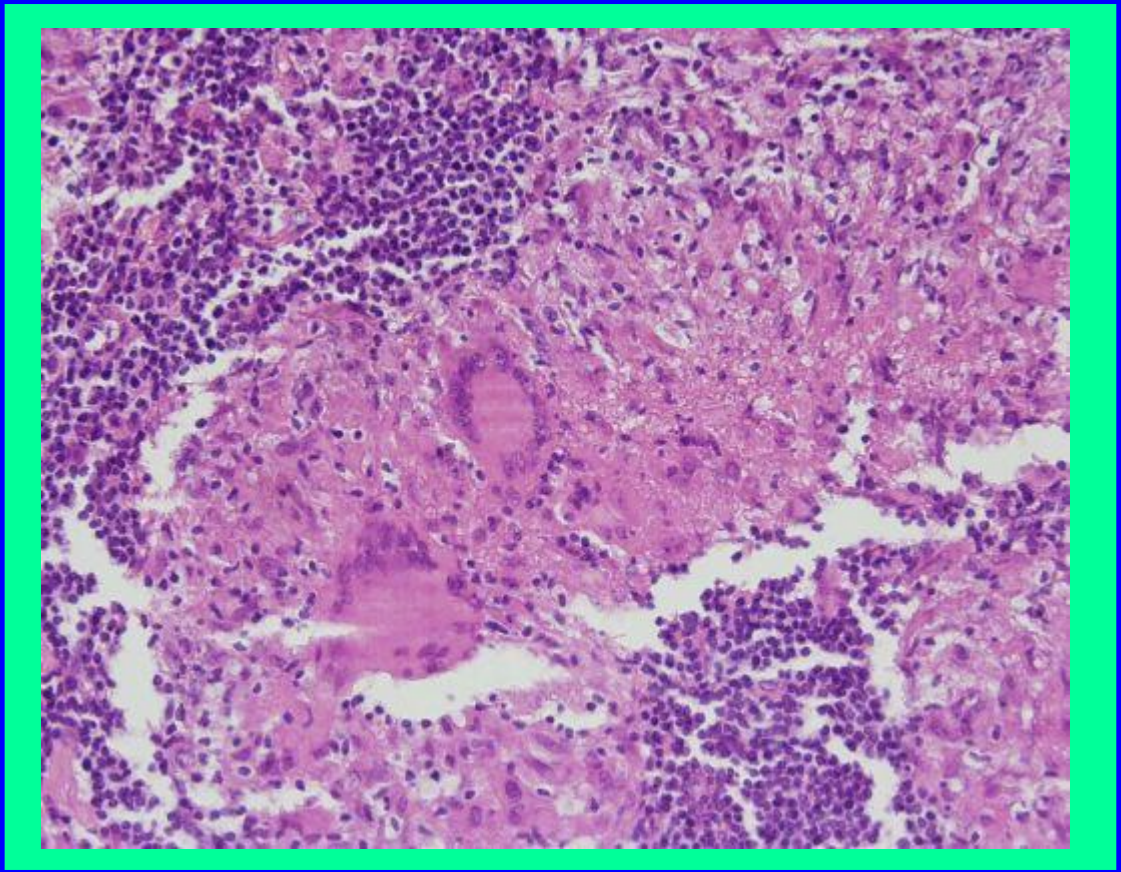




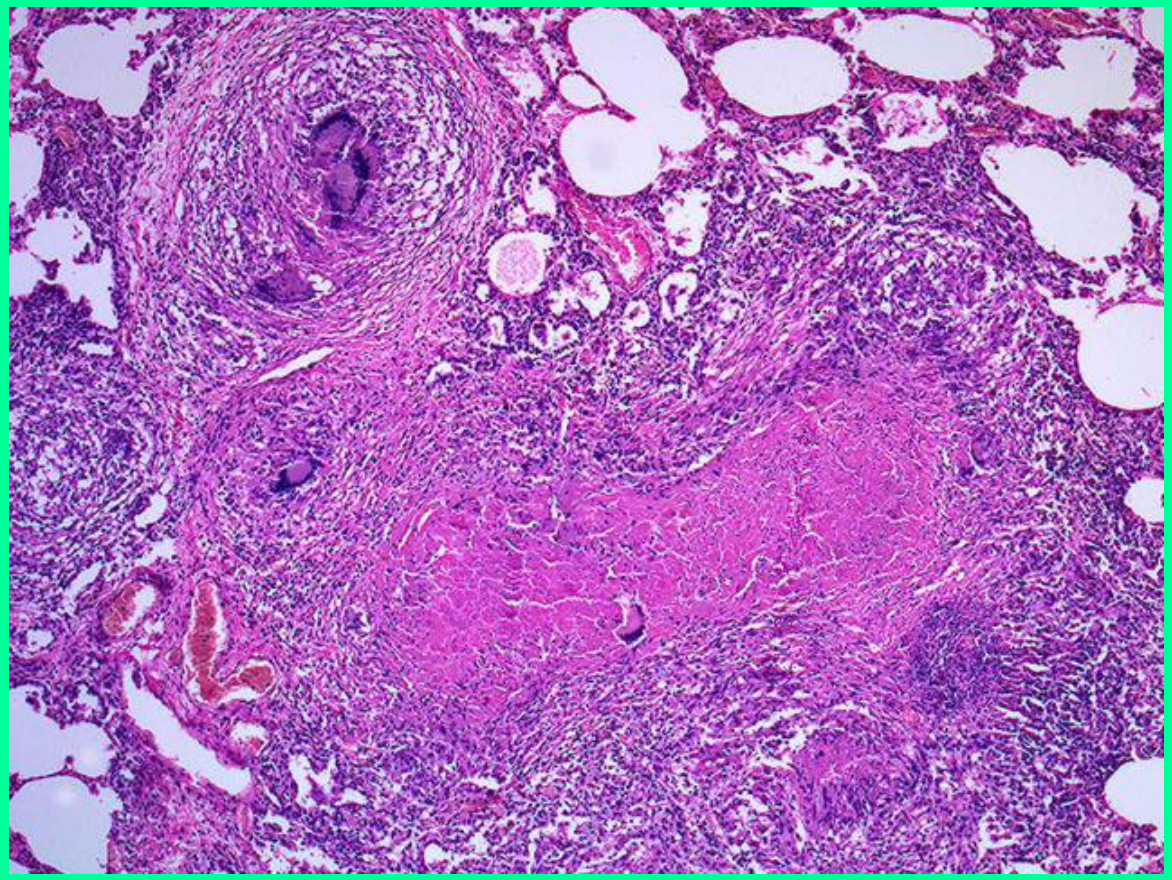
Granuloma pulmonar na sarcoidose



Granuloma na tuberculose



Granuloma miliar na tuberculose

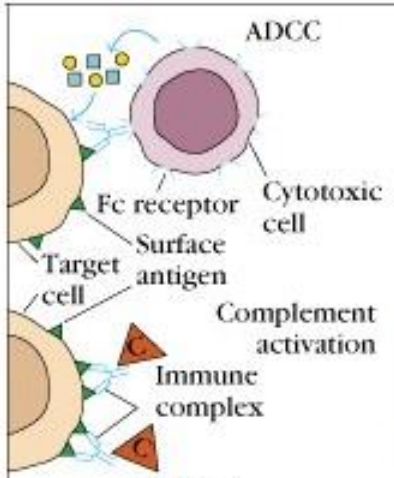
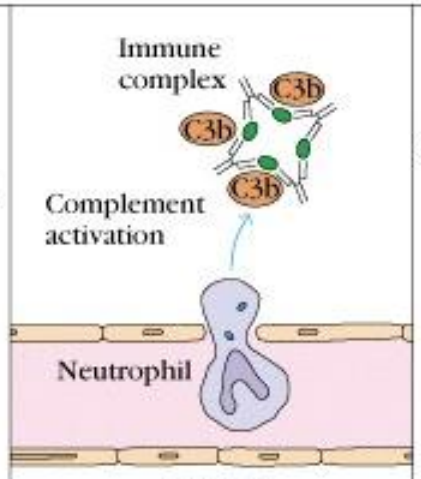
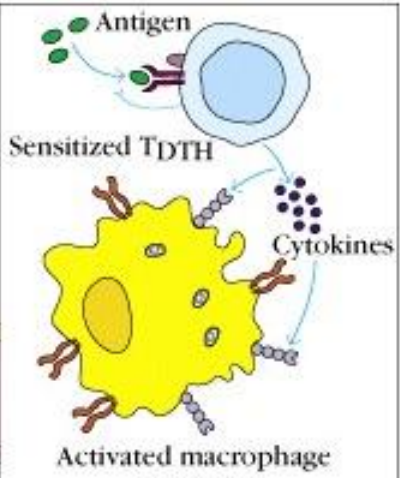


Associations of HLA genotype with susceptibility to autoimmune disease

Disease	HLA allele	Relative risk	Sex ratio (♀:♂)
Ankylosing spondylitis	B27	87.4	0.3
Acute anterior uveitis	B27	10.04	<0.5
Goodpasture's syndrome	DR2	15.9	~1
Multiple sclerosis	DR2	4.8	10
Graves' disease	DR3	3.7	4–5
Myasthenia gravis	DR3	2.5	~1
Systemic lupus erythematosus	DR3	5.8	10–20
Insulin-dependent diabetes mellitus	DR3 and DR4	3.2	~1
Rheumatoid arthritis	DR4	4.2	3
Pemphigus vulgaris	DR4	14.4	~1
Hashimoto's thyroiditis	DR5	3.2	4–5

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Mecanismos envolvidos na patologia /lesão

 **Type II**	 **Type III**	 **Type IV**
IgG-Mediated Cytotoxic Hypersensitivity	Immune Complex-Mediated Hypersensitivity	Cell-Mediated Hypersensitivity
Ab directed against cell surface antigens meditates cell destruction via complement activation or ADCC	Ag-Ab complexes deposited in various tissues induce complement activation and an ensuing inflammatory response mediated by massive infiltration of neutrophils	Sensitized T_{DTH} cells release cytokines that activate macrophages or T_C cells which mediate direct cellular damage
Typical manifestations include blood transfusion reactions, erythroblastosis fetalis, and autoimmune hemolytic anemia	Typical manifestations include localized Arthus reaction and generalized reactions such as serum sickness, necrotizing vasculitis, glomerulnephritis, rheumatoid arthritis, and systemic lupus erythematosus	Typical manifestations include contact dermatitis, tubercular lesions and graft rejection

Aspecto das três principais reações cutâneas



Tipo I
(imediata)

Tipo III
Arthus
(5-12 h)

Tipo IV
(tardia)

Aspecto das três principais reações cutâneas



Tipo I
(imediata)

Tipo III Arthus
(5-12 h)

Tipo IV
(tardia)

Reação tuberculínica (PPD): Indução
Th1/Th17/ Th2

Hipersensibilidade tipo IV ou hipersensibilidade Tardia

hipersensibilidade mediada por células CD8+

Exemplo: vírus da Hepatite

Hipersensibilidade tipo I ou imediata (Atopia)

RESPOSTA A ALÉRGENOS MEDIADA PELA IgE

Enquanto que a maior parte dos indivíduos não faz anticorpos (ou faz preferencialmente anticorpos das classes IgM ou IgG) frente a alguns antígenos inócuos,

..... uma fatia importante da população (que chega a 20% nos USA) faz níveis elevados de IgE

Os antígenos que desencadeiam hipersensibilidade imediata são denominados de ALÉRGENOS

ALÉRGENOS

- Pólen de plantas
- Drogas (penicilina, sulfonamidas, anestésicos locais, salicilatos)
- Alimentos (nozes, frutos do mar, ovos, ervilha, leite)
- Produtos de artrópodos (veneno de vespa, abelha...., picada de formiga, fezes de acaros)
- Esporos de fungos
- Pelos de animais

Elementos-chave da Hipersensibilidade tipo I

-IgE

-Th2

-Basófilos, Mastócitos

- Eosinófilos

- receptores $Fc_{\varepsilon}R1$

Os antígenos que desencadeiam hipersensibilidade imediata são denominados de ALÉRGENOS

E os quadros clínicos de hipersensibilidade tipo I são denominados de alergias

(correto ou incorreto?)

Febre do feno e Rinite crônica

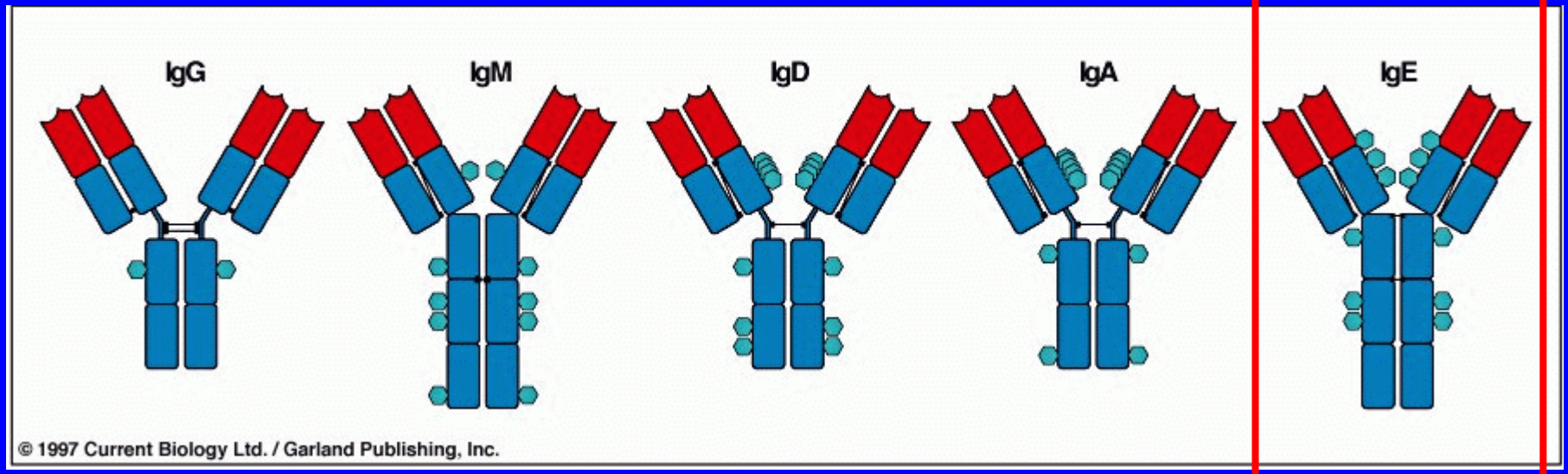
Asma Brônquica

Alergia a alimentos, drogas

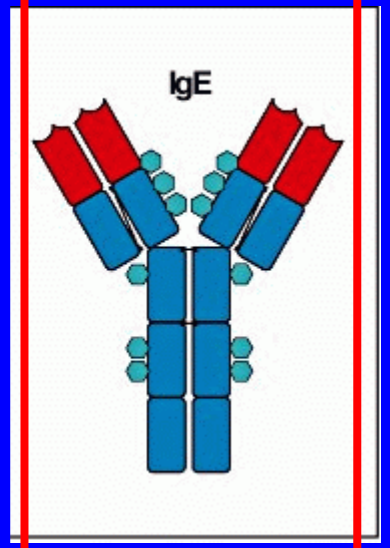
Eczema atópico

Choque anafilático (sistêmico)

MECANISMOS EFETORES MEDIADOS PELA IgE


IgG
IgM
IgD
IgA
IgE
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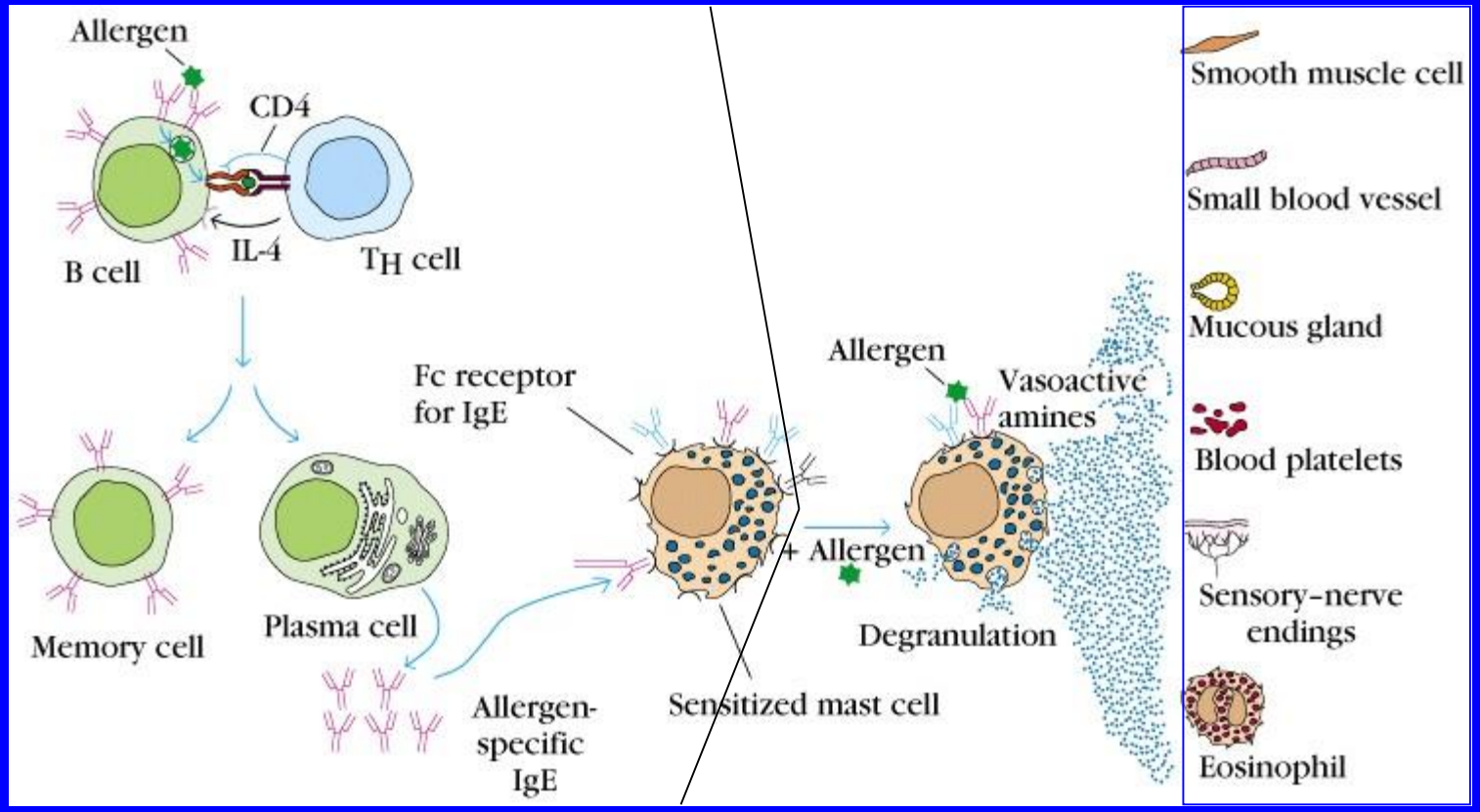
Qual a clínica do primeiro contato com o alérgeno (antígeno indutor da alergia) ?



Não há hipersensibilidade tipo I no primeiro contato com um alérgeno, mas somente em contatos posteriores

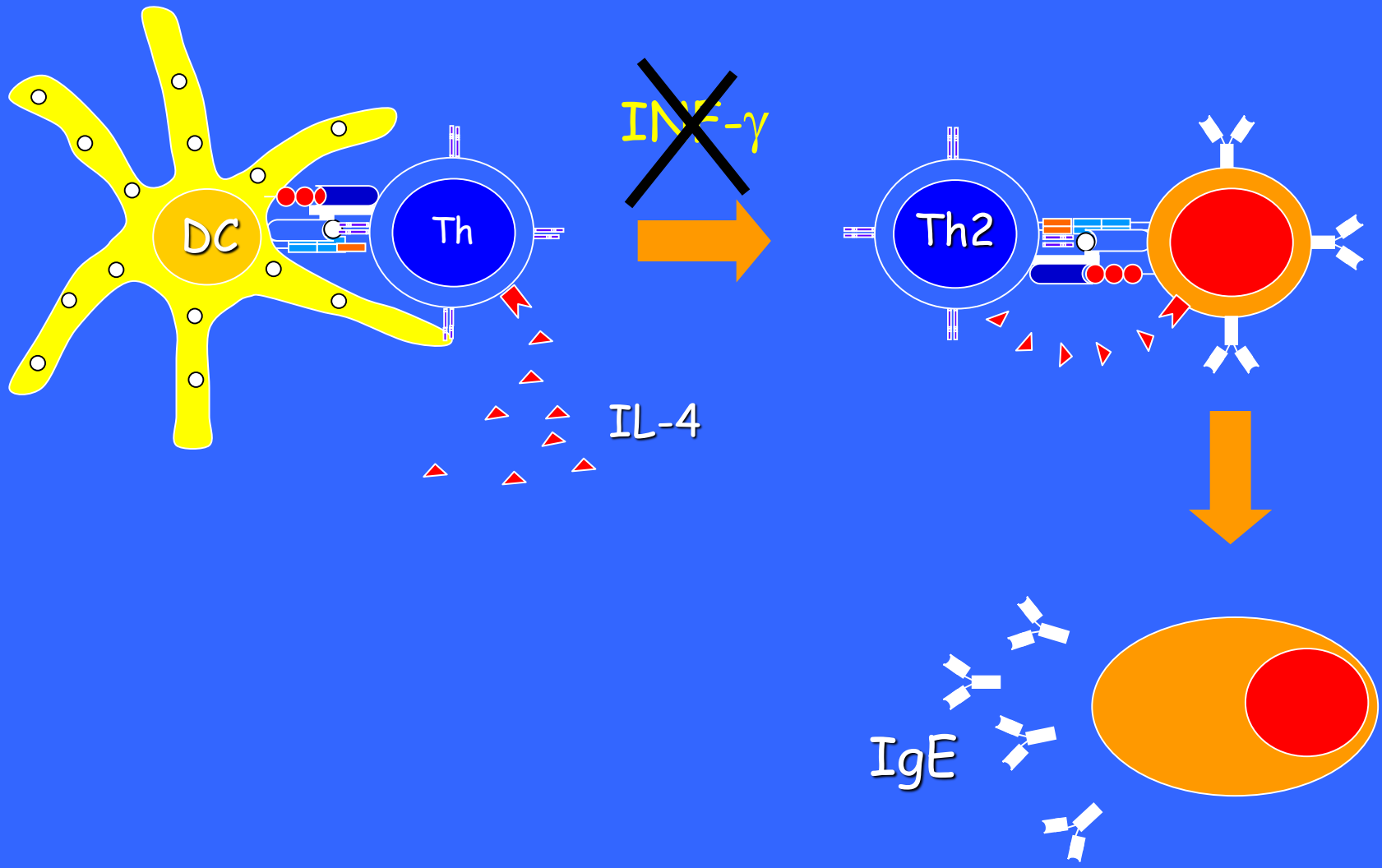
Sensibilização e desencadeamento de Alergia tipo I

ALVOS INDIRETOS DA AÇÃO EFETORA DA IgE



Sensibilização Desencadeamento

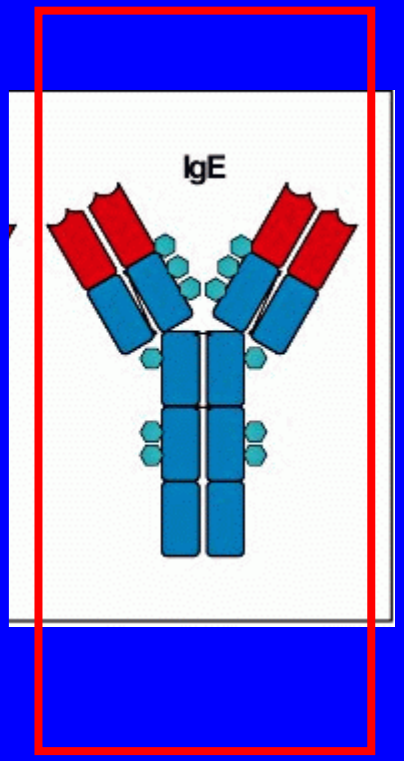
Como são produzidos os anticorpos da classe IgE?



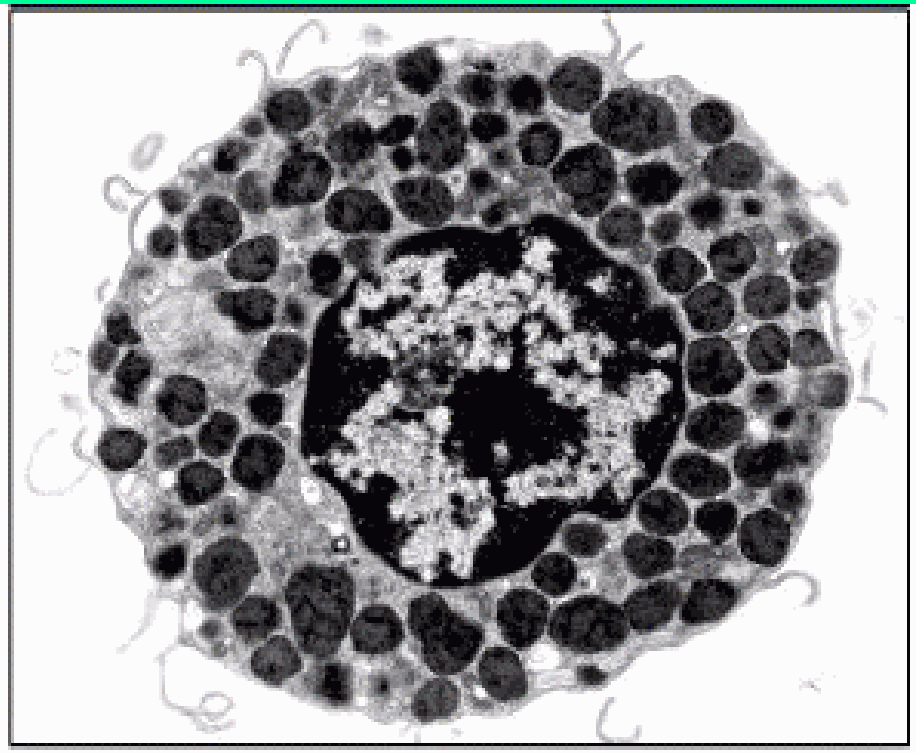
Prausnitz e Kustner descreveram que a inoculação de soro de um paciente alérgico na pele de um indivíduo não alérgico (teste) com posterior injeção de alérgeno provoca rubor e inchaço

- Ivan Mota descobriu que este efeito do soro é devido à degranulação de mastócitos e é mediado por uma globulina

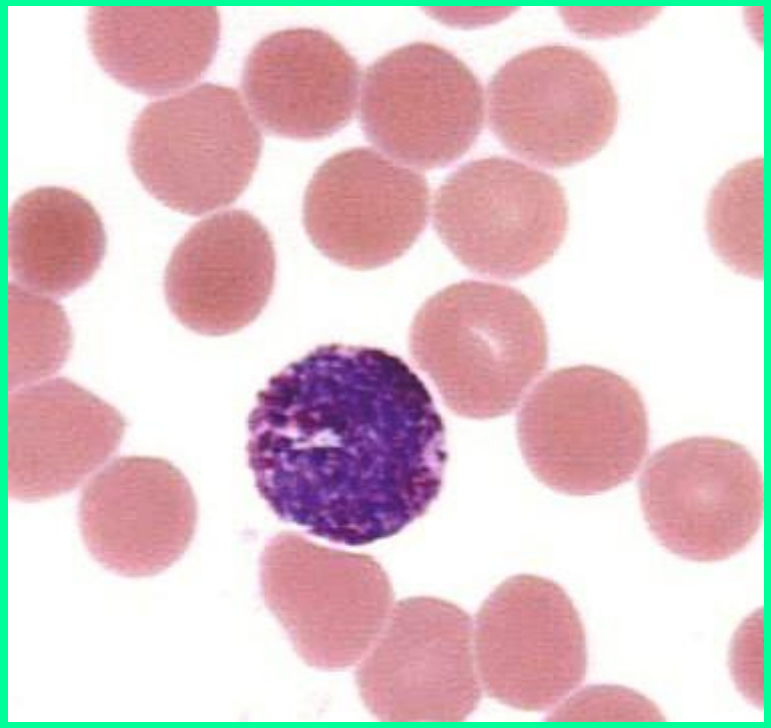
- A IgE foi descoberta por K. e T. Ishizaka



Mastócito

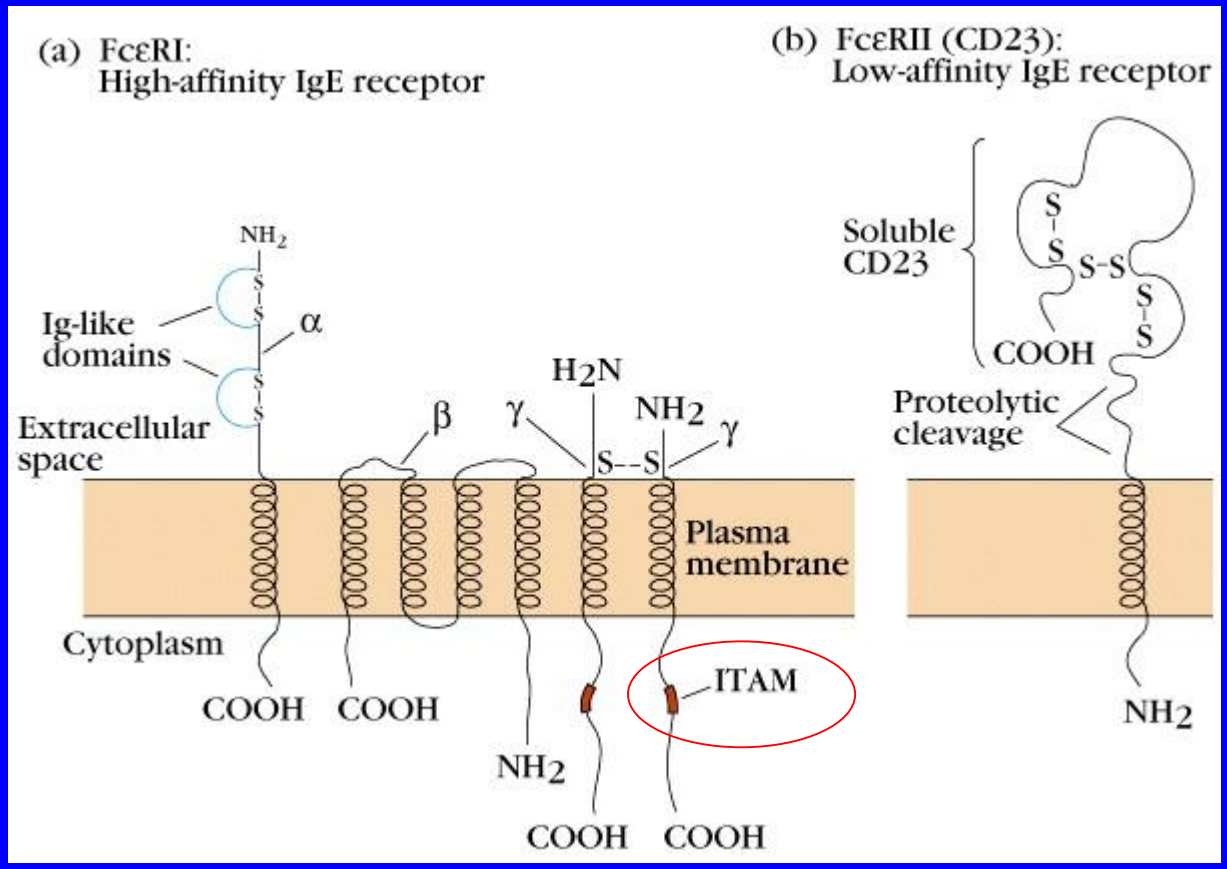


Basófilo

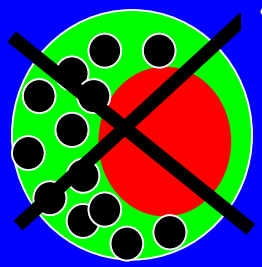


Os basófilos e mastócitos têm receptores para IgE

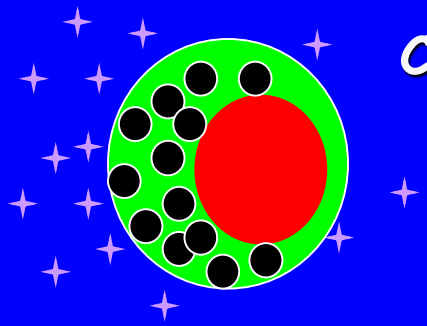
FcεRs



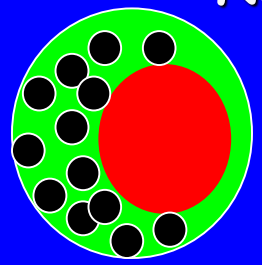
DEGRANULAÇÃO



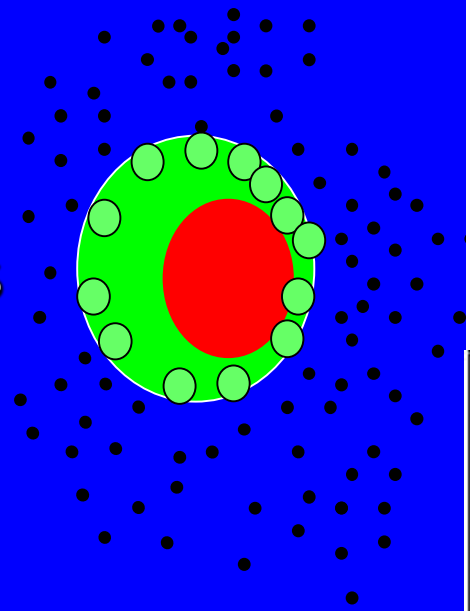
TRAUMA

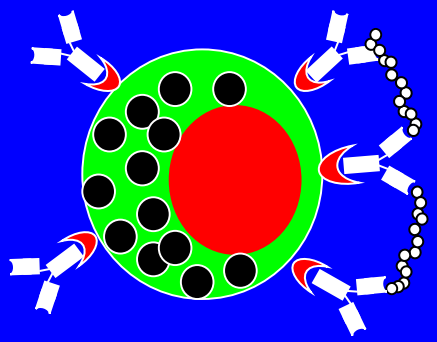


C3a, C5a

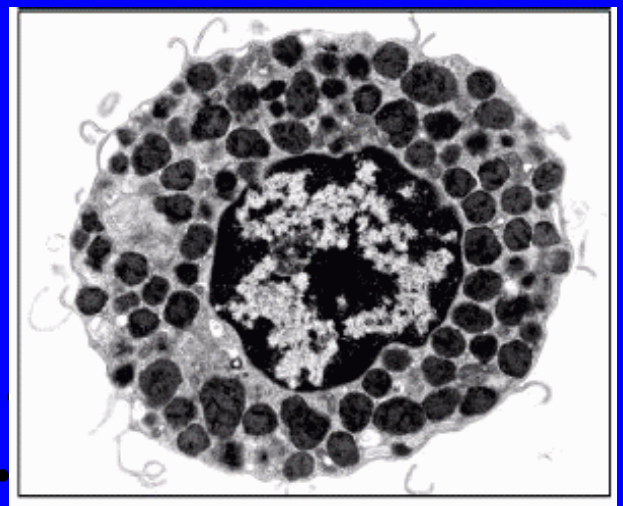


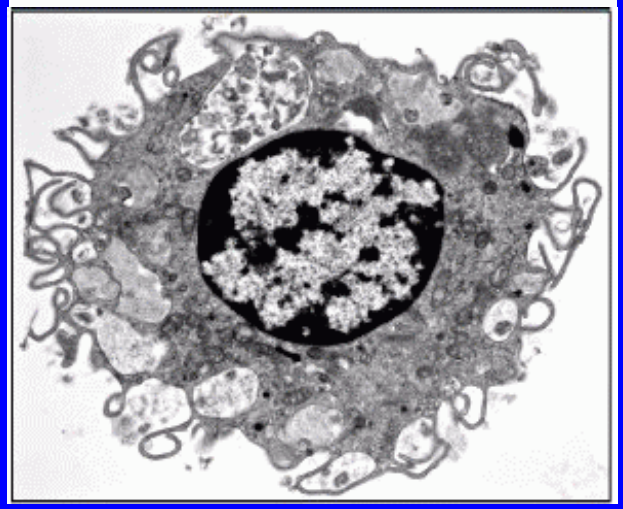
NEUROPEPTÍDEOS



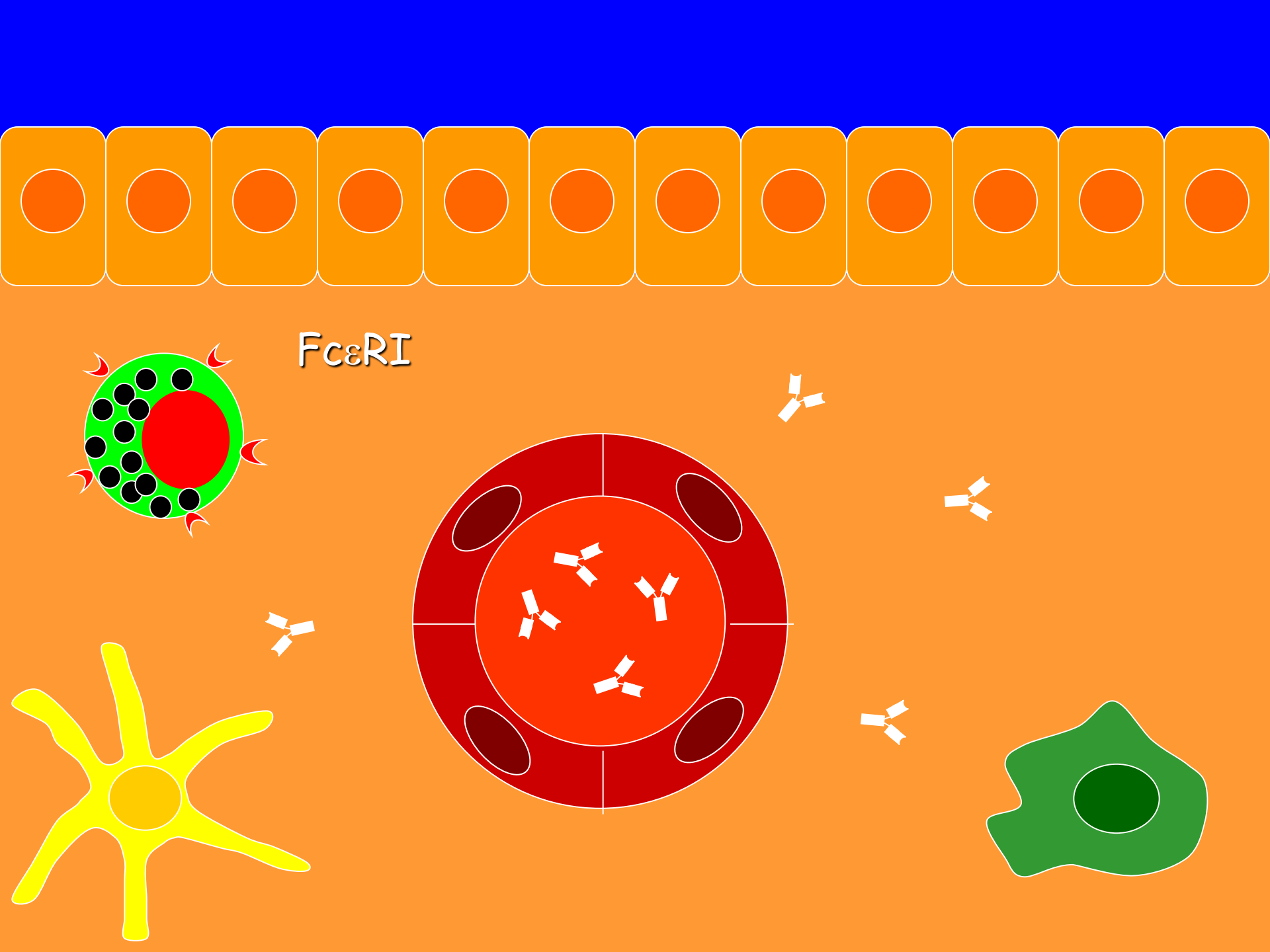


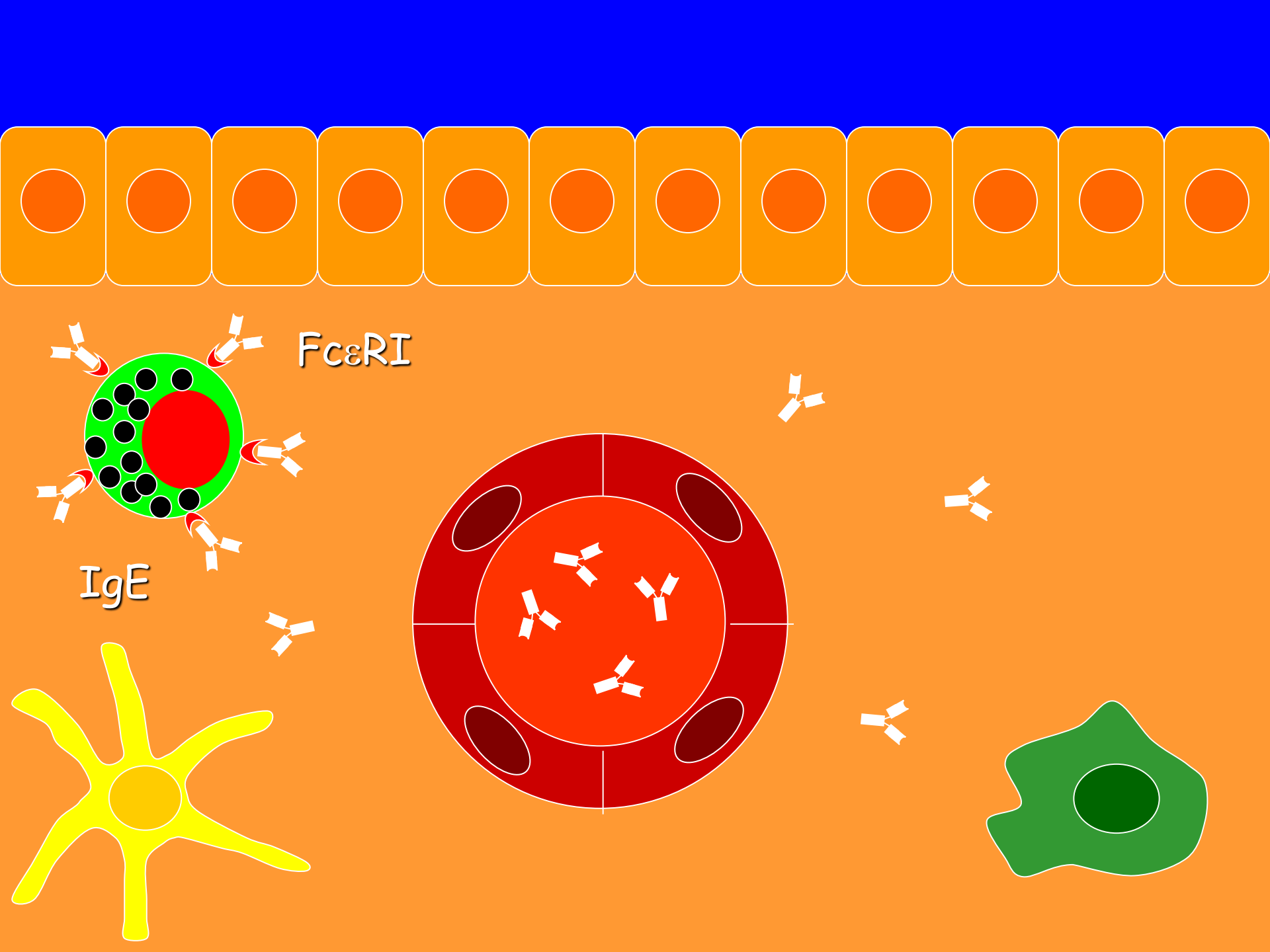
AGRUPAMENTO DA IgE





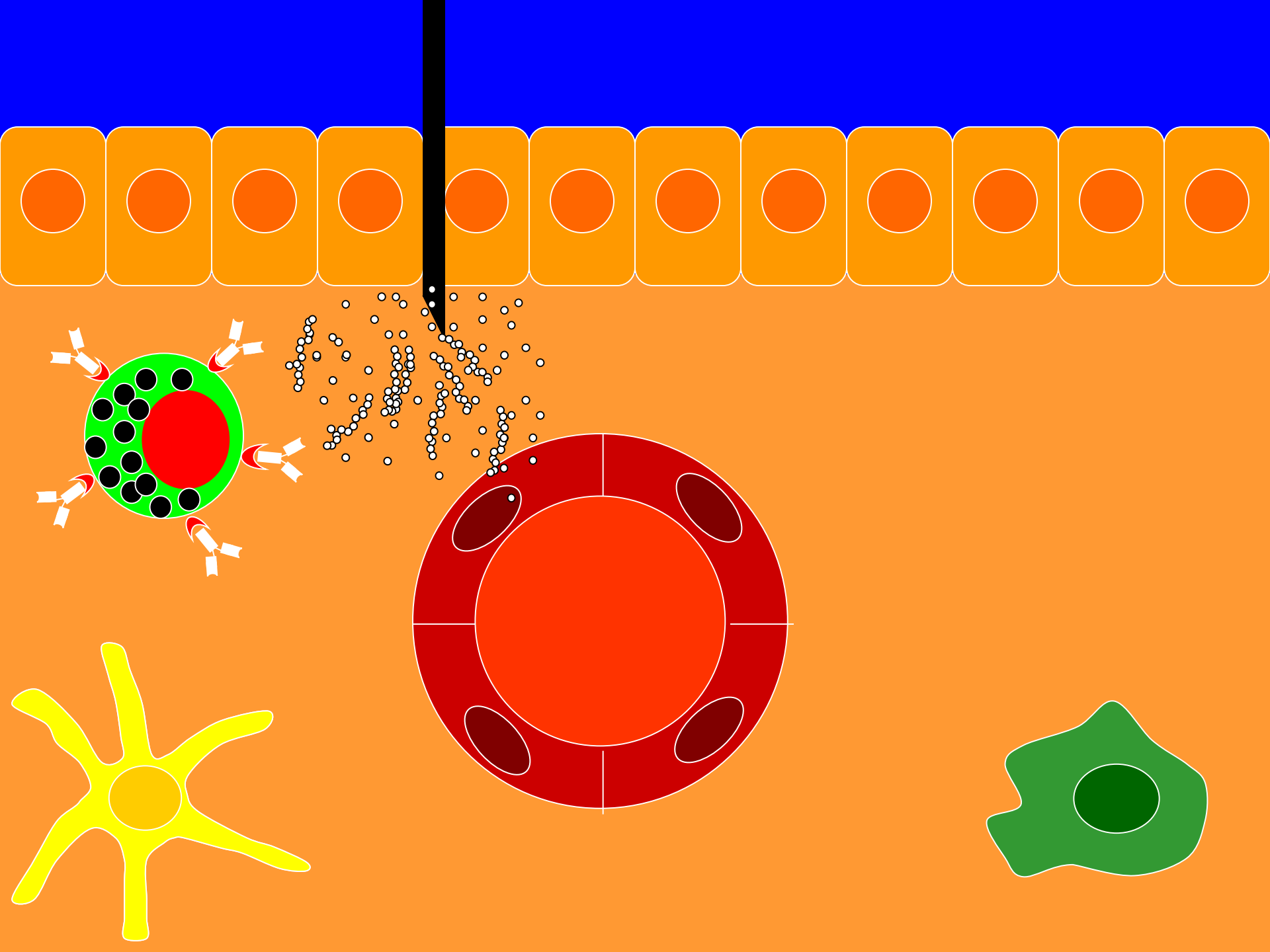
Enquanto a IgE no soro dura somente 2-3 dias, na superfície do mastócito dura semanas

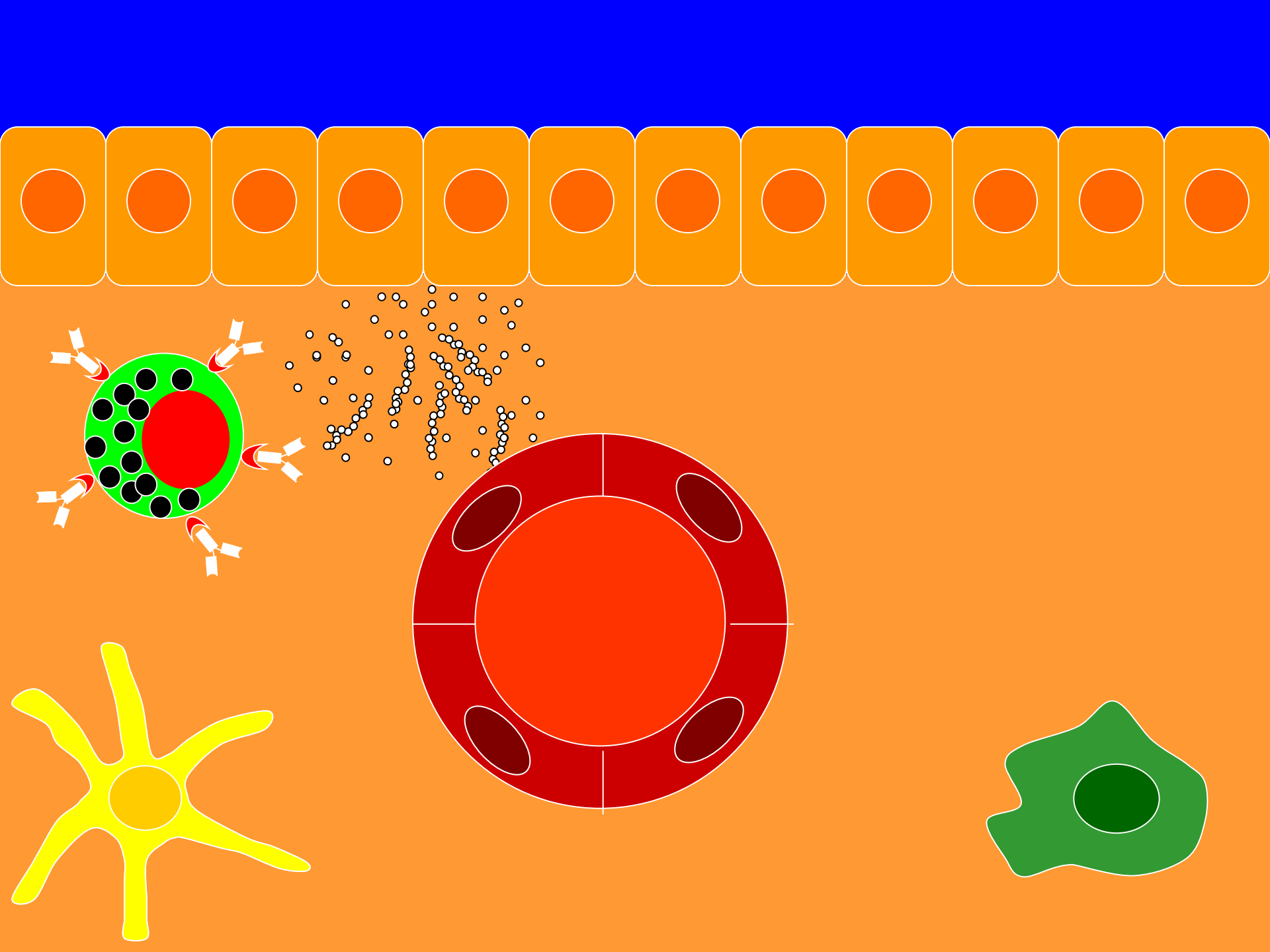

FcεRI

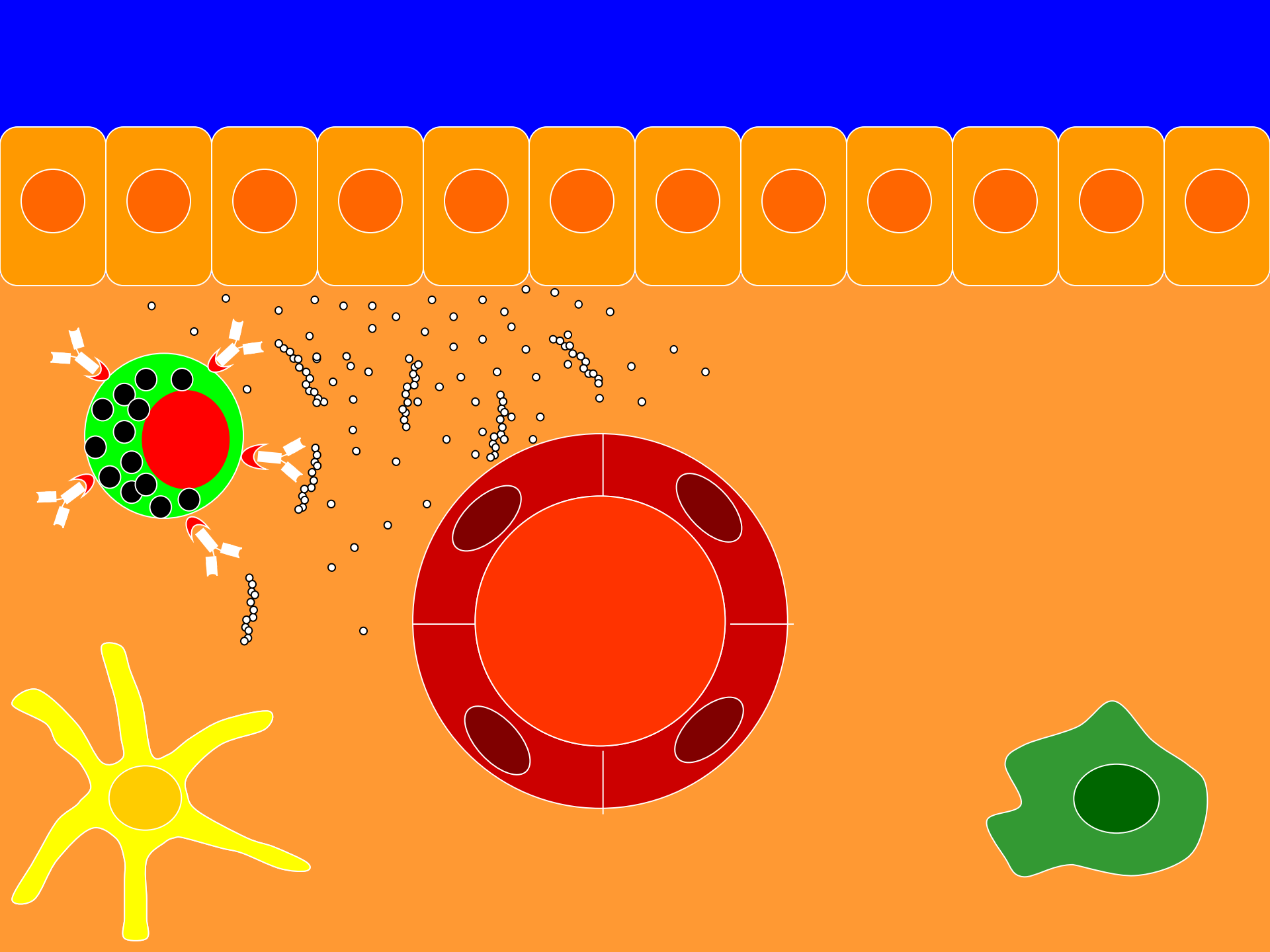

FcεRI
IgE

Desencadeamento da Resposta

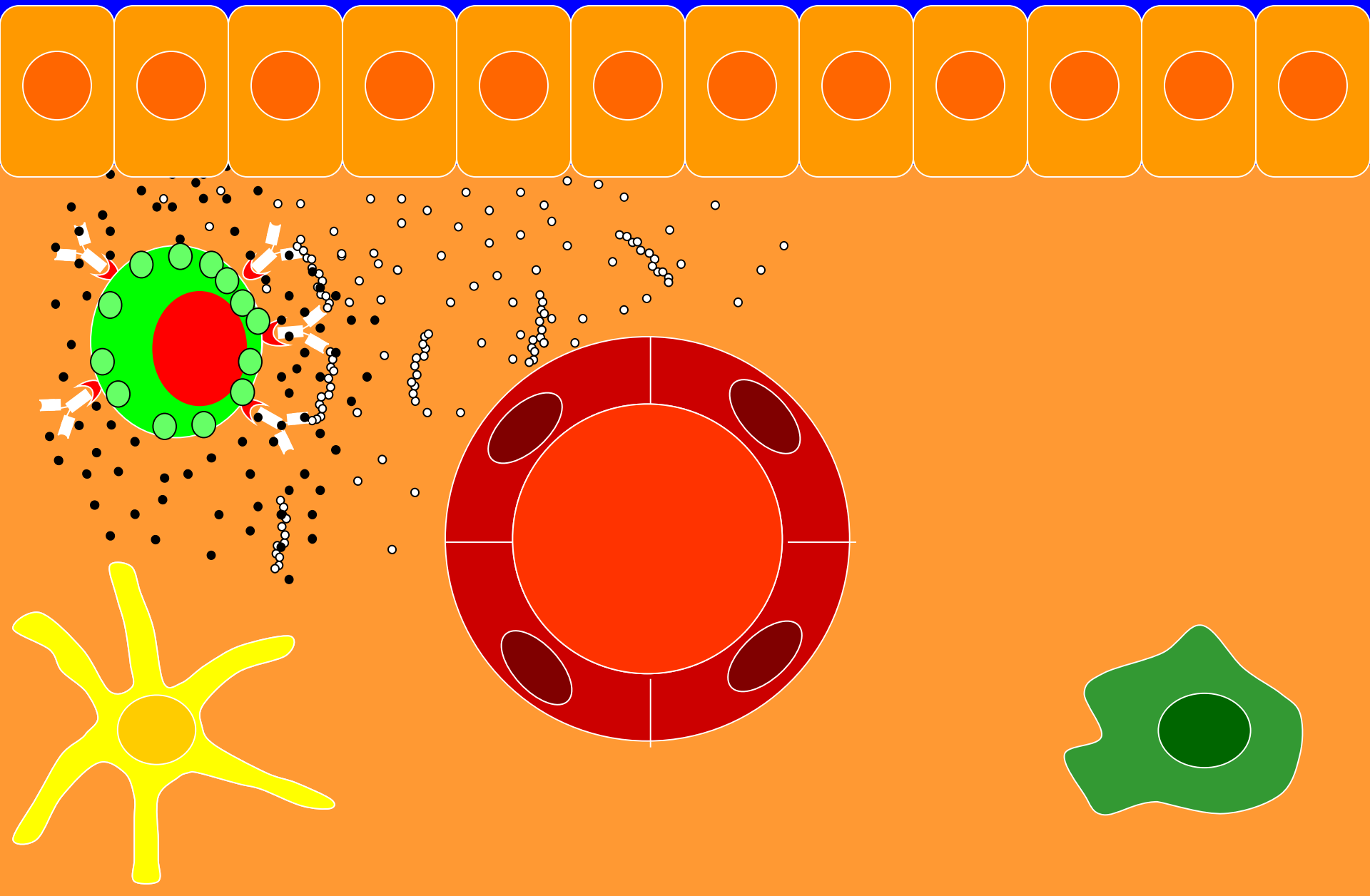
Reação imediata

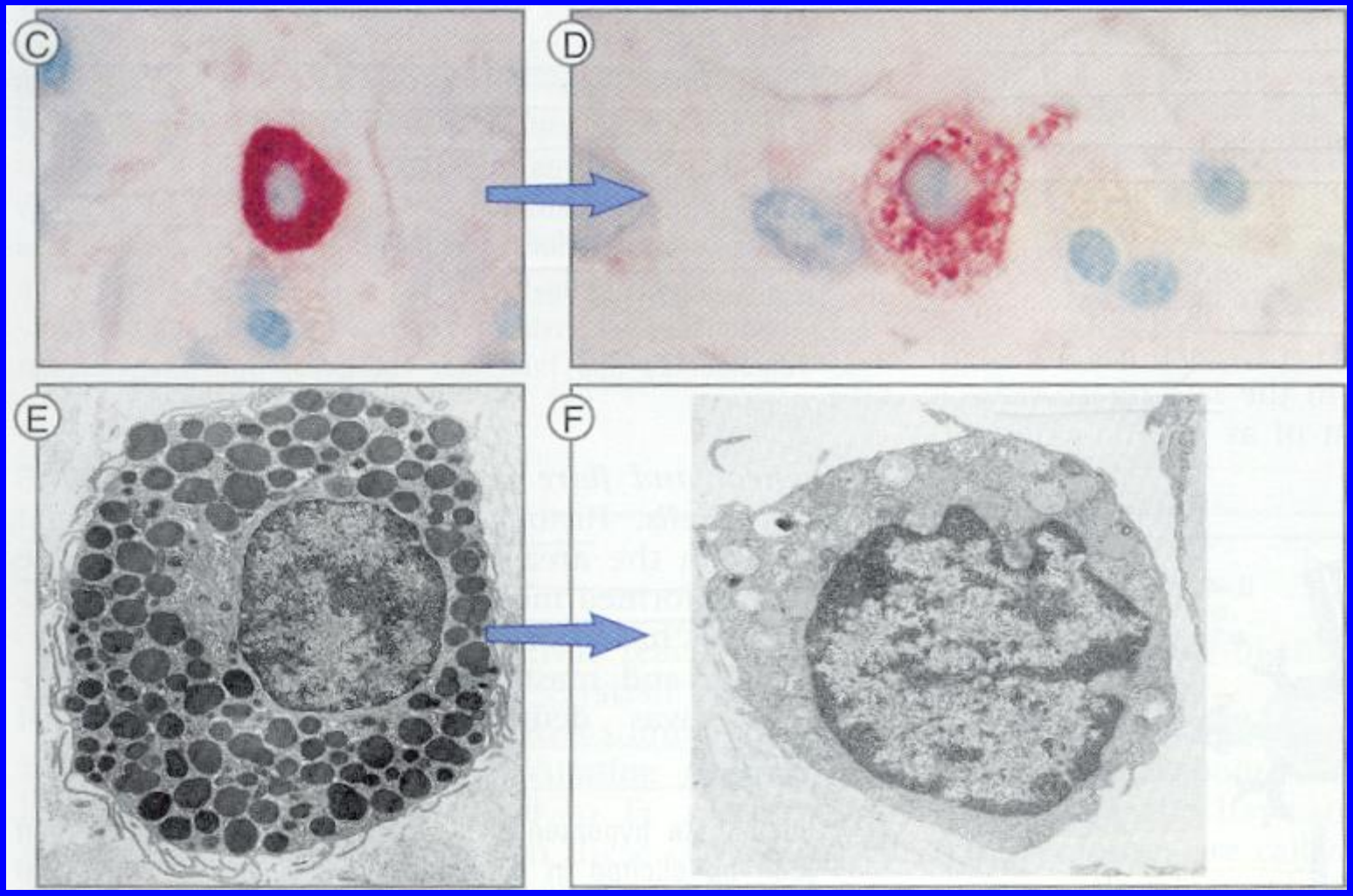


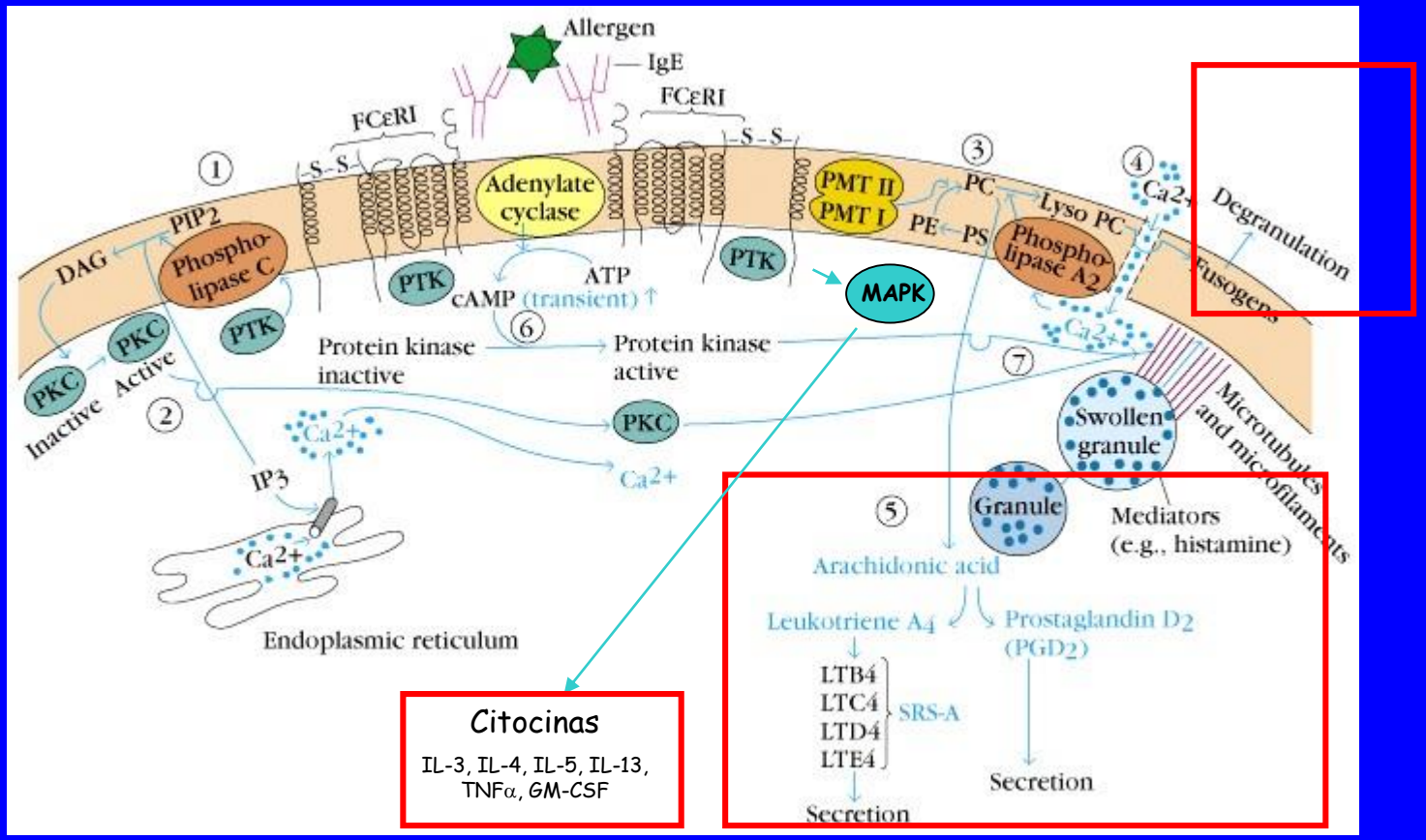




Reação imediata




C
D
E
F


Allergen
IgE
FCεRI
FCεRI
S–S
S–S
1
Adenylate cyclase
PMT II
PMT I
3
PC
Lyso PC
PE
PS
Phospho-lipase A2
4
Ca2+
Degranulation
Fusogens
PIP2
DAG
Phospho-lipase C
PTK
PTK
PTK
ATP
cAMP (transient) ↑
MAPK
6
Protein kinase inactive
Protein kinase active
PKC
Active
PKC
Inactive
2
PKC
7
Ca2+
Ca2+
IP3
Ca2+
Ca2+
Endoplasmic reticulum
Swollen granule
Microtubules and microfilaments
5
Granule
Mediators (e.g., histamine)
Arachidonic acid
Leukotriene A4
Prostaglandin D2 (PGD2)
LTB4
LTC4
LTD4
LTE4
SRS-A
Secretion
Secretion
Citocinas
IL-3, IL-4, IL-5, IL-13, TNFα, GM-CSF

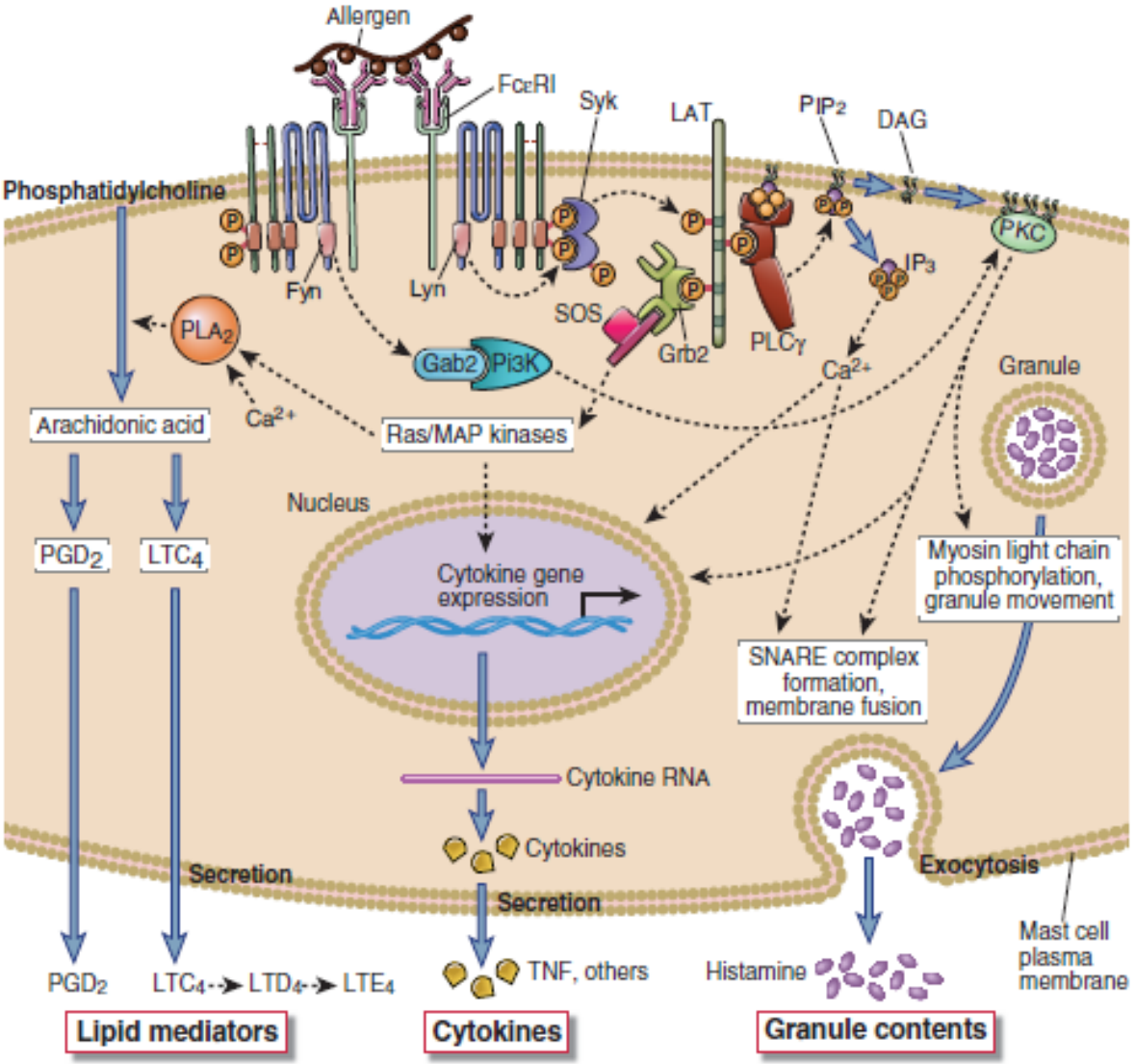


FIGURE 19–5 Biochemical events of mast cell activation. Cross-linking of bound IgE by antigen is thought to activate protein tyrosine kinases (Syk and Lyn), which in turn cause activation of a MAP kinase cascade and a phosphatidylinositol-specific phospholipase C (PI-PLCγ). PI-PLCγ catalyzes the release of IP3 and DAG from membrane PIP2. IP3 causes release of intracellular calcium from the endoplasmic reticulum. Calcium and DAG activate PKC, which phosphorylates substrates such as myosin light chain protein and thereby leads to the degradation and release of preformed mediators. Calcium and MAP kinases combine to activate the enzyme cytosolic phospholipase A_2 (PLA_2), which initiates the synthesis of lipid mediators, including prostaglandin D_2 (PGD_2) and leukotriene C_4 (LTC_4).

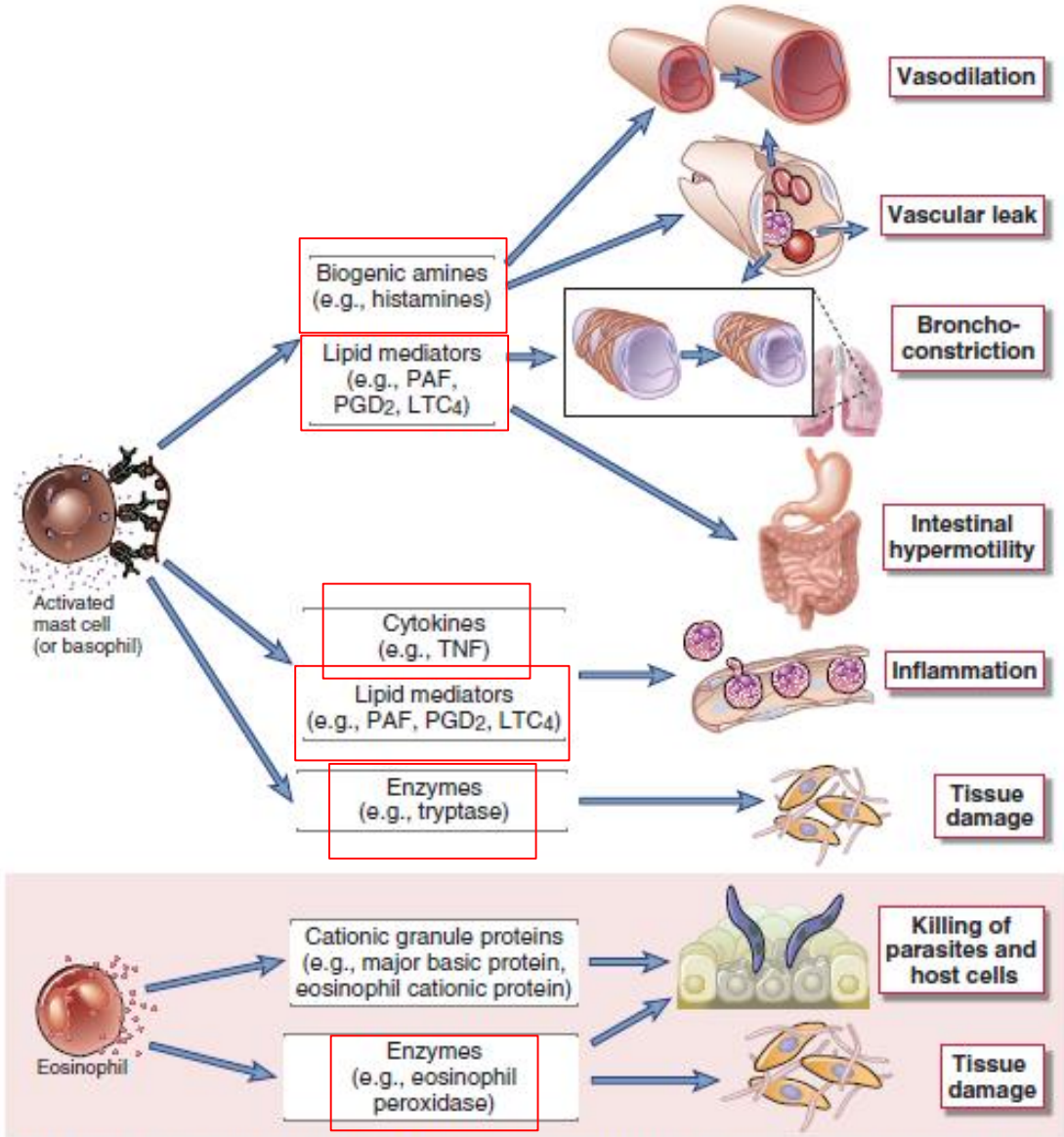


FIGURE 19–6 Biologic effects of mediators of immediate hypersensitivity. Mast cells and basophil mediators include biogenic amines and enzymes stored preformed in granules as well as cytokines and lipid mediators, which are largely newly synthesized on cell activation. The biogenic amines and lipid mediators induce vascular leakage, bronchoconstriction, and intestinal hypermotility, all components of the immediate response. Cytokines and lipid mediators contribute to inflammation, which is part of the late-phase reaction. Enzymes probably contribute to tissue damage. Activated eosinophils release preformed cationic proteins as well as enzymes that are toxic to parasites and host cells. Some eosinophil granule enzymes probably contribute to tissue damage in chronic allergic diseases.

vasodilatação

aumento de permeabilidade

boncoconstrição

aumento da motilidade intestinal

inflamação

dano tissular

dano tissular

dano tissular

Reação Imediata (minutos)

Fase Tardia (horas e pode perdurar até dias)

Reação Imediata (minutos)

Fase Tardia (horas e pode perdurar até dias)

ASMA BRôNQUICA

Reação Imediata

-histamina, PGs e LTs

vasodilatação e edema broncoconstrição

Fase Tardia

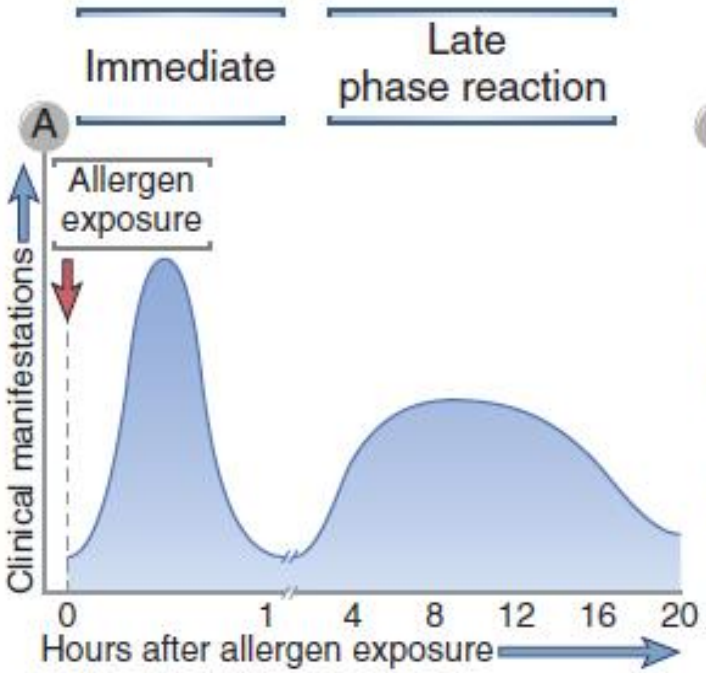
- IL-4, IL-5, IL-6, TNFα, ECF, PAF

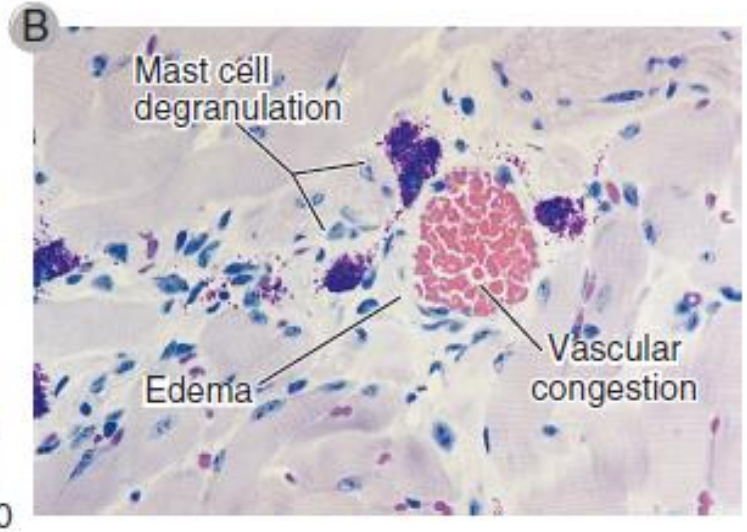
↓

Recrutamento de eosinófilos e neutrófilos

↓

Lesão tissular





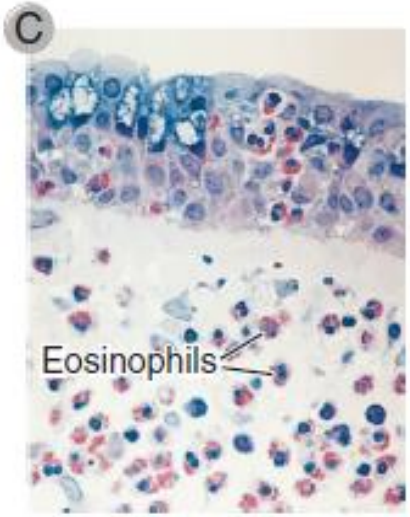
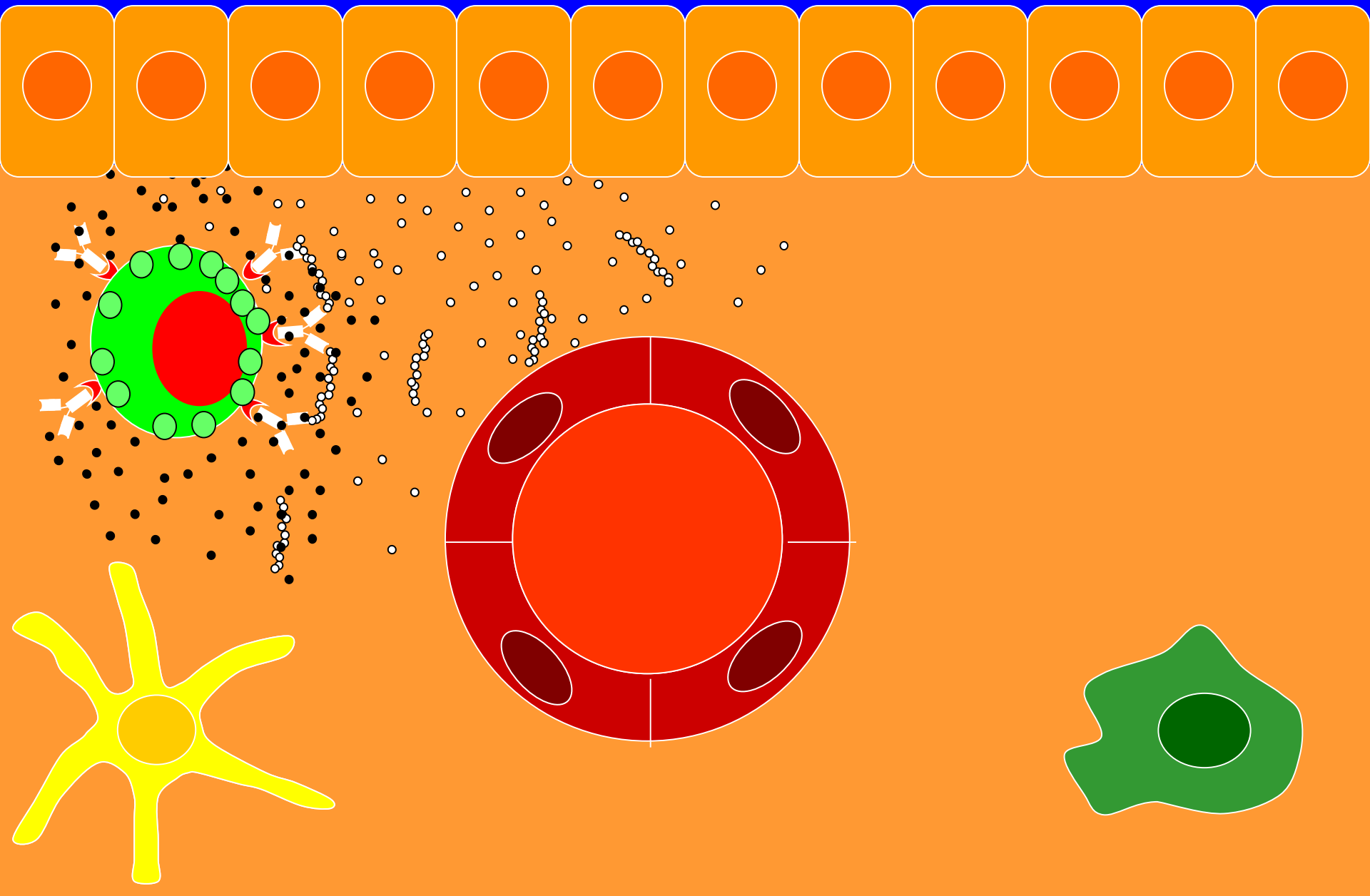
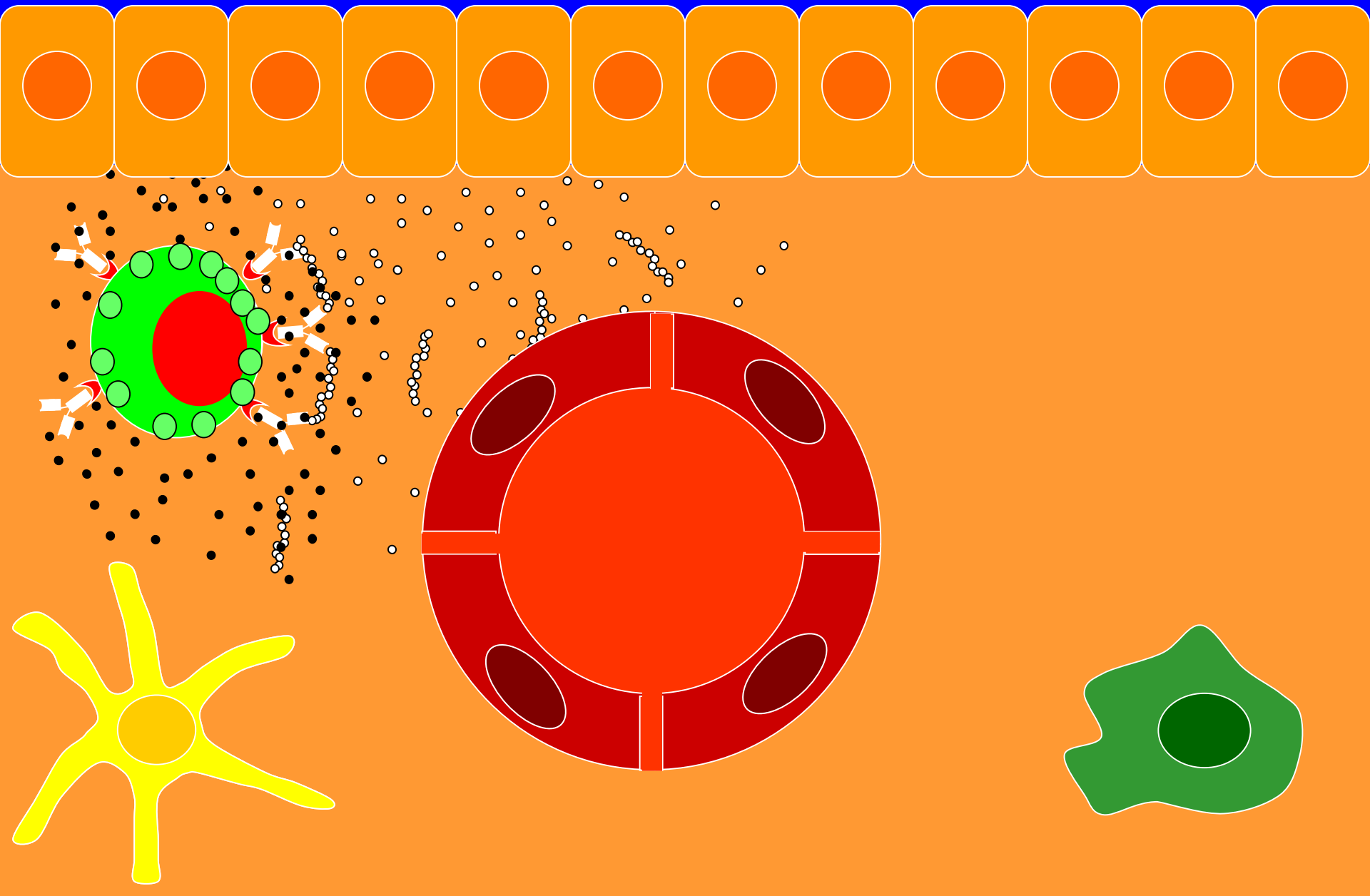


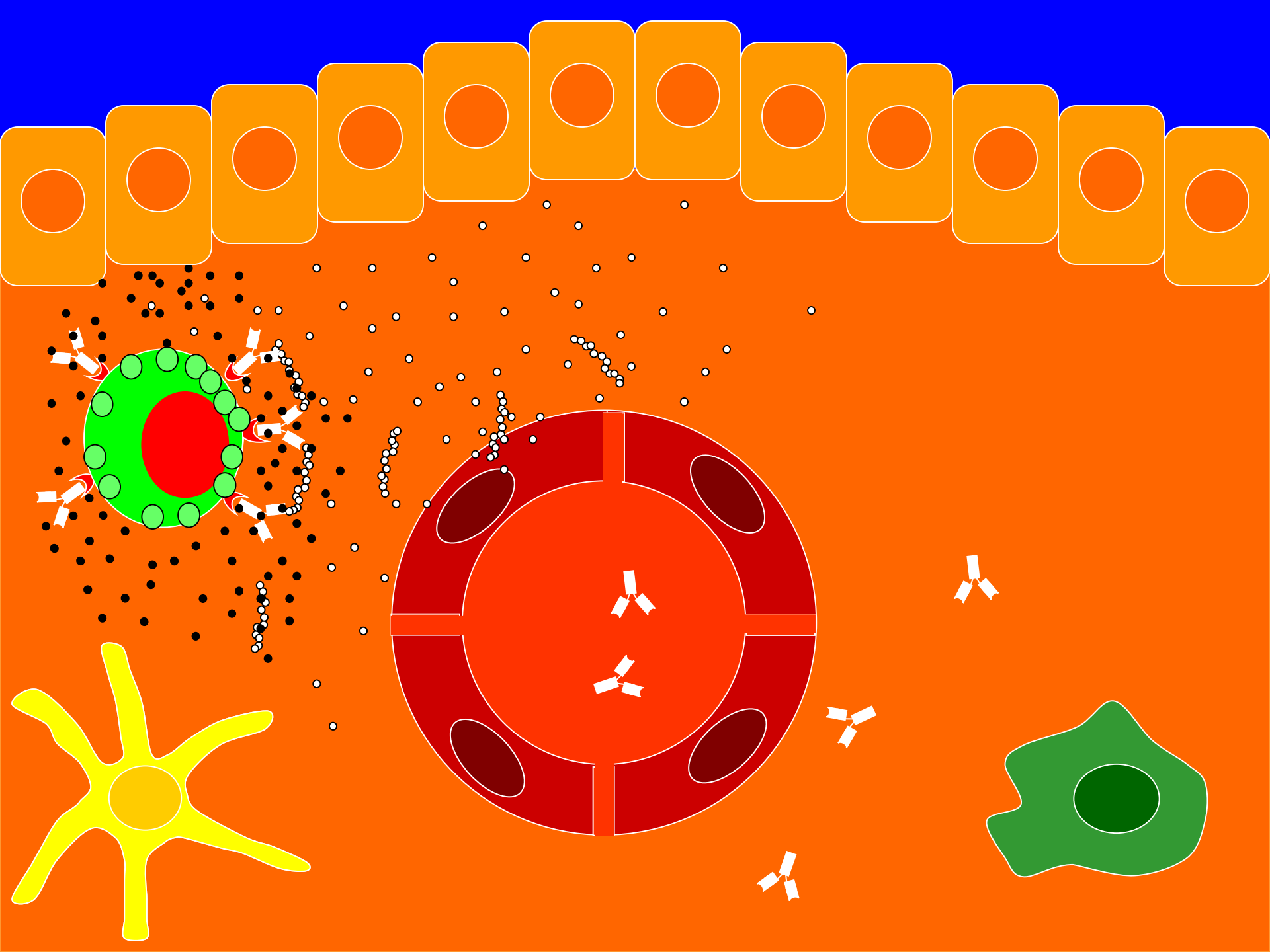
FIGURE 19–7 The immediate and late-phase reactions of allergy. A, Kinetics. The immediate vascular and smooth muscle reaction to allergen develops within minutes after challenge (allergen exposure in a previously sensitized individual), and the late-phase reaction develops 2 to 24 hours later. **B, C,** Morphology. The immediate reaction **(B)** is characterized by vasodilation, congestion, and edema, and the late-phase reaction **(C)** is characterized by an inflammatory infiltrate rich in eosinophils, neutrophils, and T cells. *(Courtesy of Dr. Daniel Friend, Department of Pathology, Brigham and Women's Hospital, Boston, Massachusetts.)*

Reação imediata

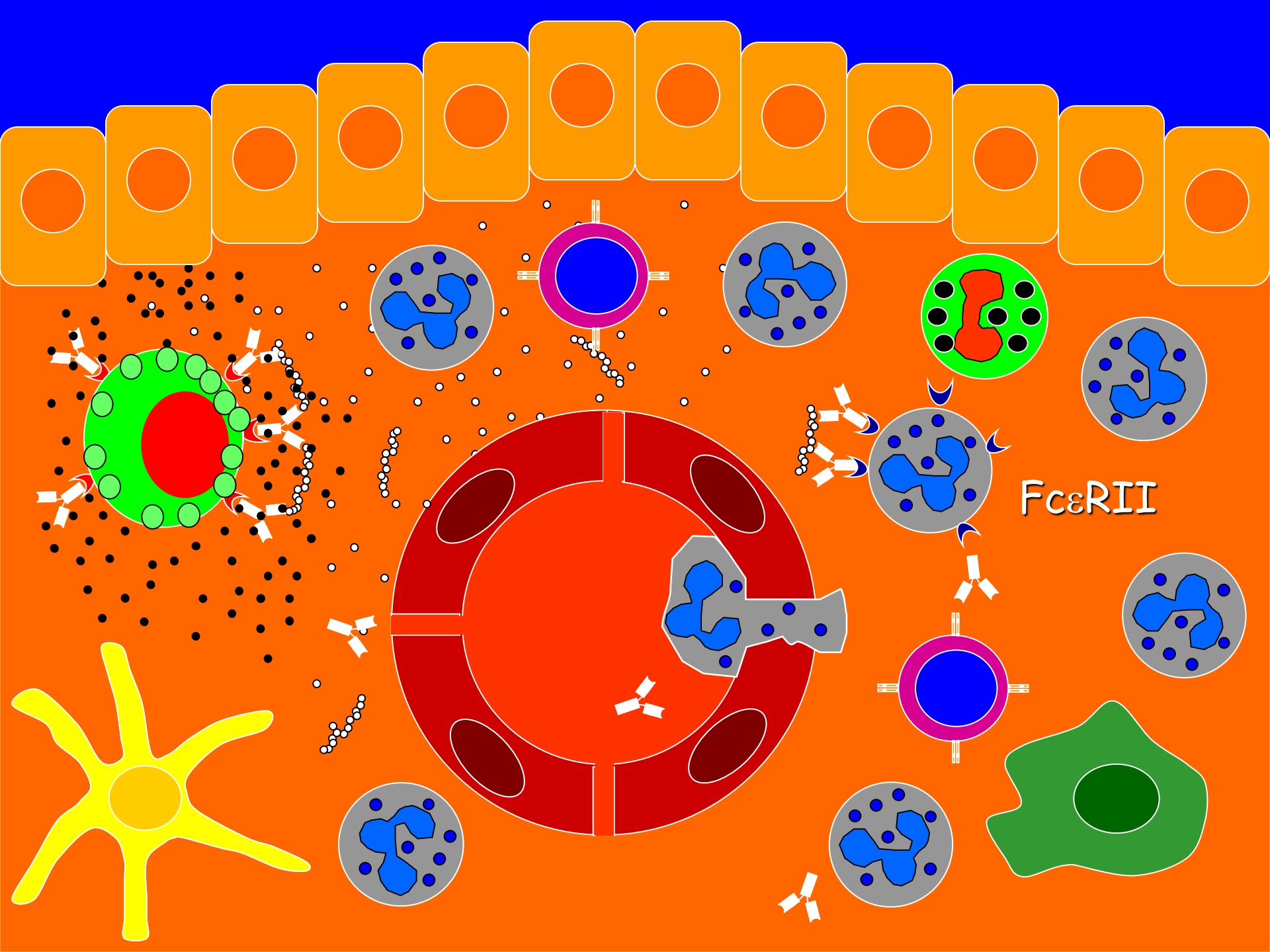


Reação imediata

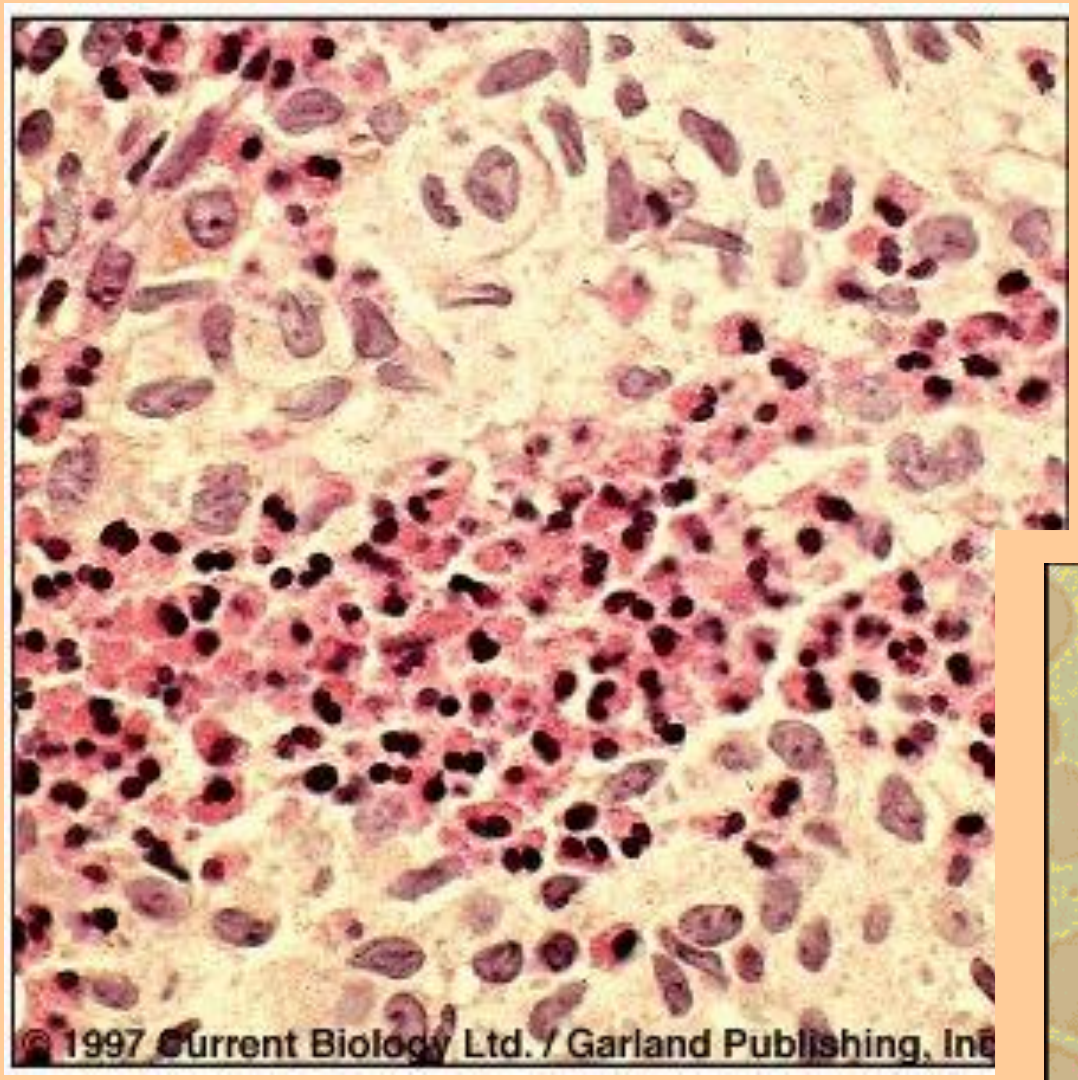


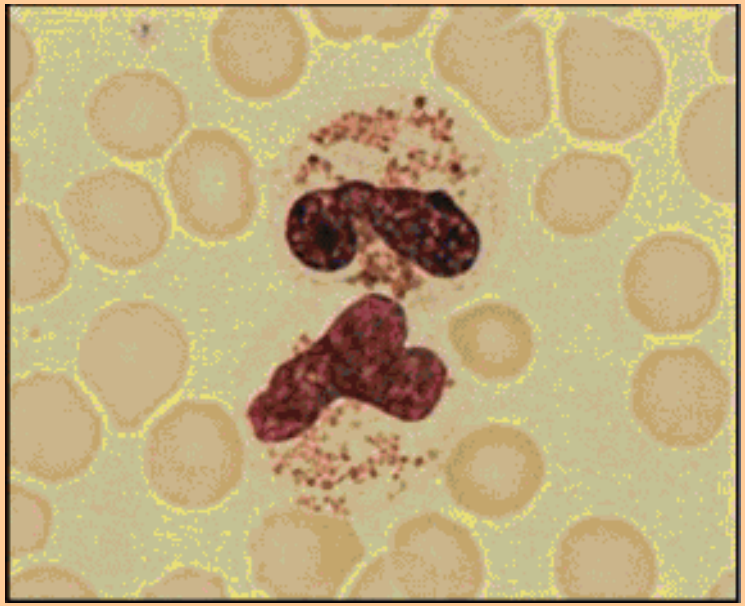


Fase tardia da resposta


FcεRII

Eosinófilos em infiltrado inflamatório e sangue





Esinófilo



Fases da alergia (hipersensibilidade tipo I)

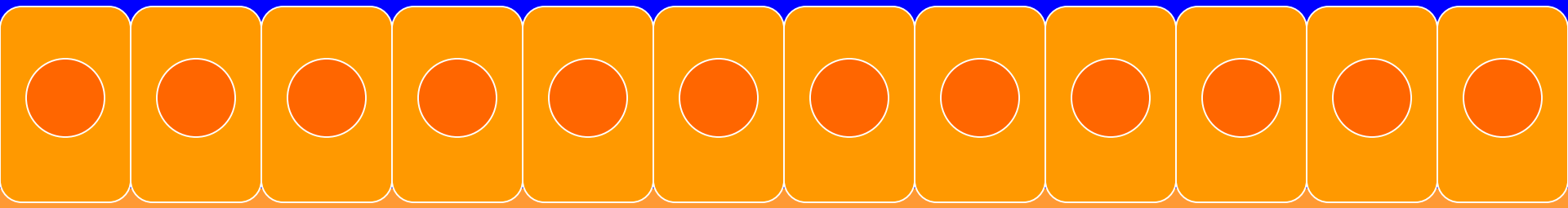


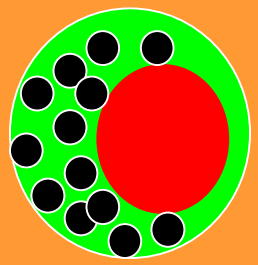
imediata

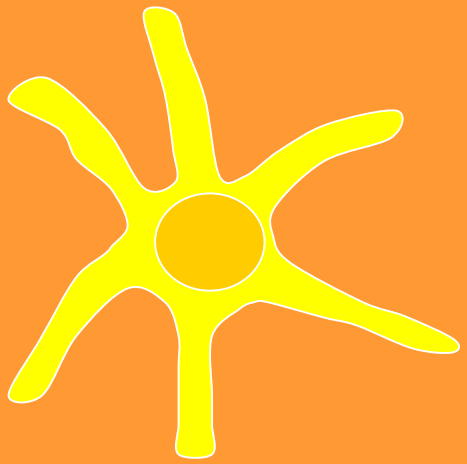


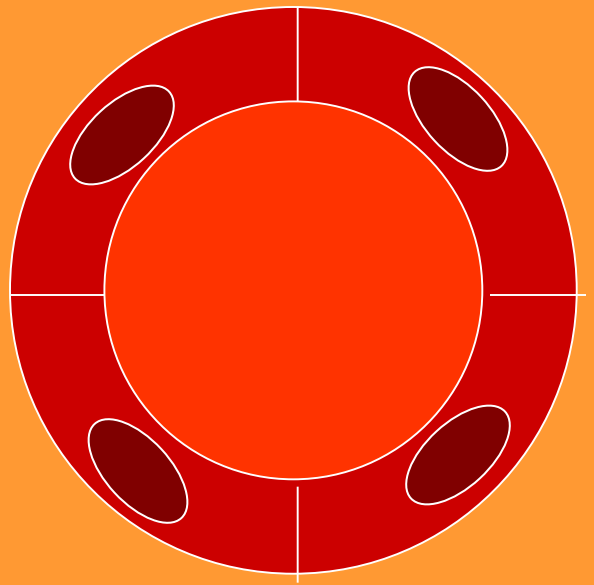
tardia

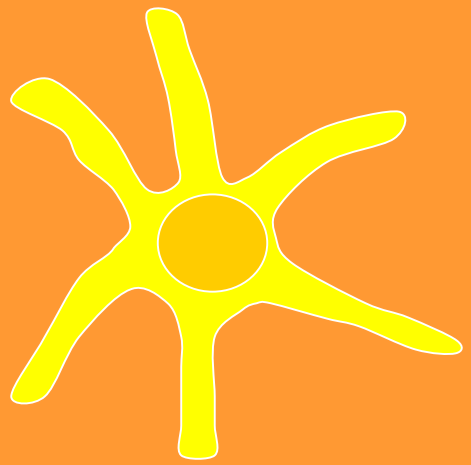
Qual é a contribuição da IgE?

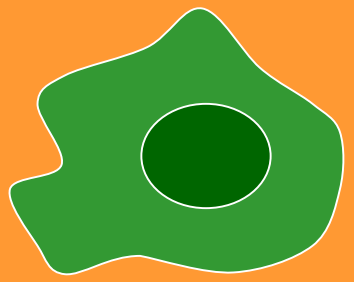


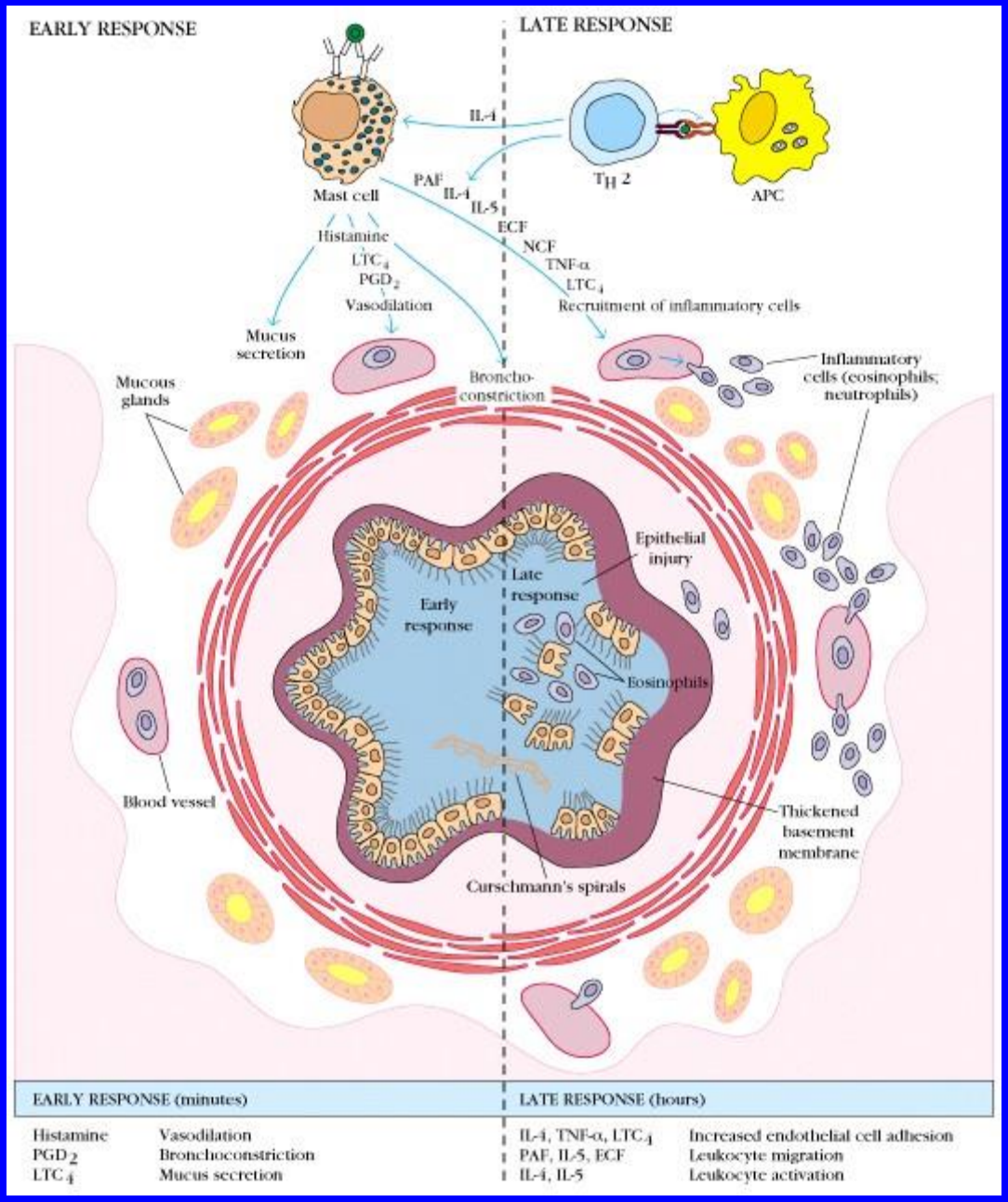




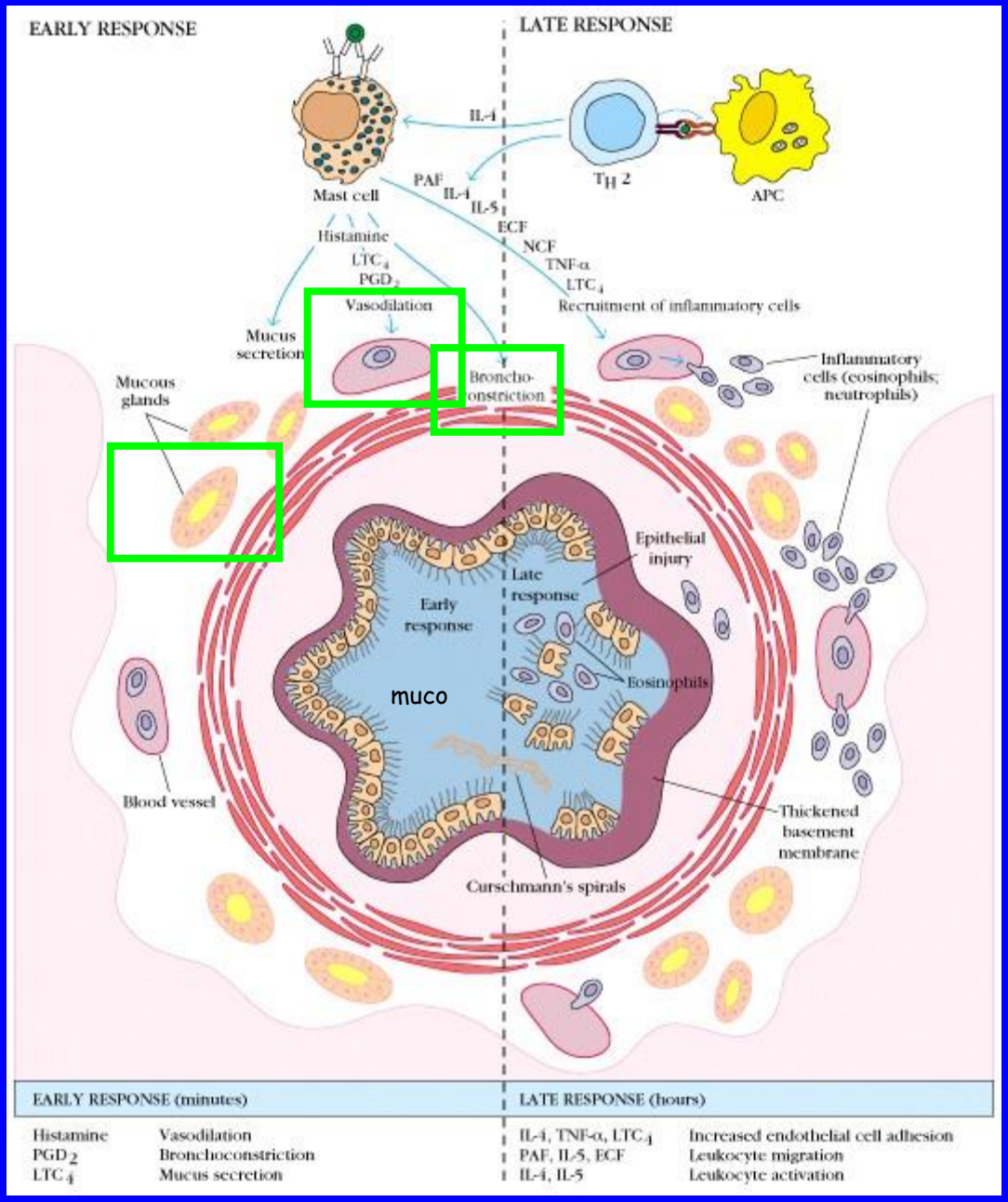


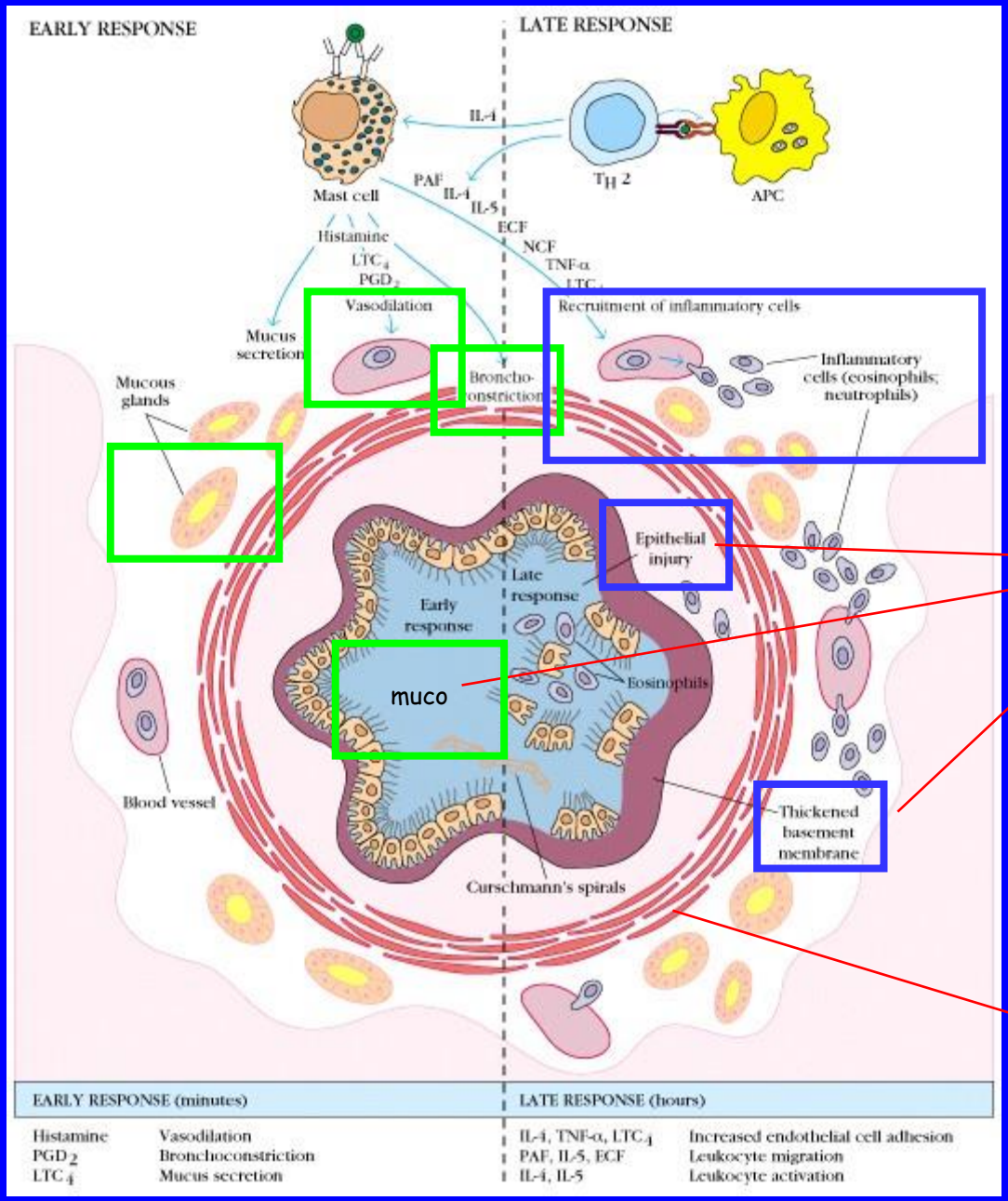






EARLY RESPONSE (minutes)		LATE RESPONSE (hours)	
Histamine	Vasodilation	IL-4, TNF-α, LTC_4	Increased endothelial cell adhesion
PGD_2	Bronchoconstriction	PAF, IL-5, ECF	Leukocyte migration
LTC_4	Mucus secretion	IL-4, IL-5	Leukocyte activation


EARLY RESPONSE
LATE RESPONSE
IL-4
Mast cell
T_H 2
APC
PAF
IL-4
IL-5
ECF
NCF
TNF-α
LTC_4
Recruitment of inflammatory cells
Histamine
LTC_4
PGD_2
Vasodilation
Mucus secretion
Broncho-constriction
Mucous glands
Inflammatory cells (eosinophils; neutrophils)
Epithelial injury
Late response
Early response
Eosinophils
muco
Blood vessel
Thickened basement membrane
Curschmann's spirals
EARLY RESPONSE (minutes)
LATE RESPONSE (hours)
Histamine
PGD_2
LTC_4
Vasodilation
Bronchoconstriction
Mucus secretion
IL-4, TNF-α, LTC_4
PAF, IL-5, ECF
IL-4, IL-5
Increased endothelial cell adhesion
Leukocyte migration
Leukocyte activation



Oclusão bronquial

Hipertrofia musculatura brônquica

Th2

-IL-4 ---------- IgE

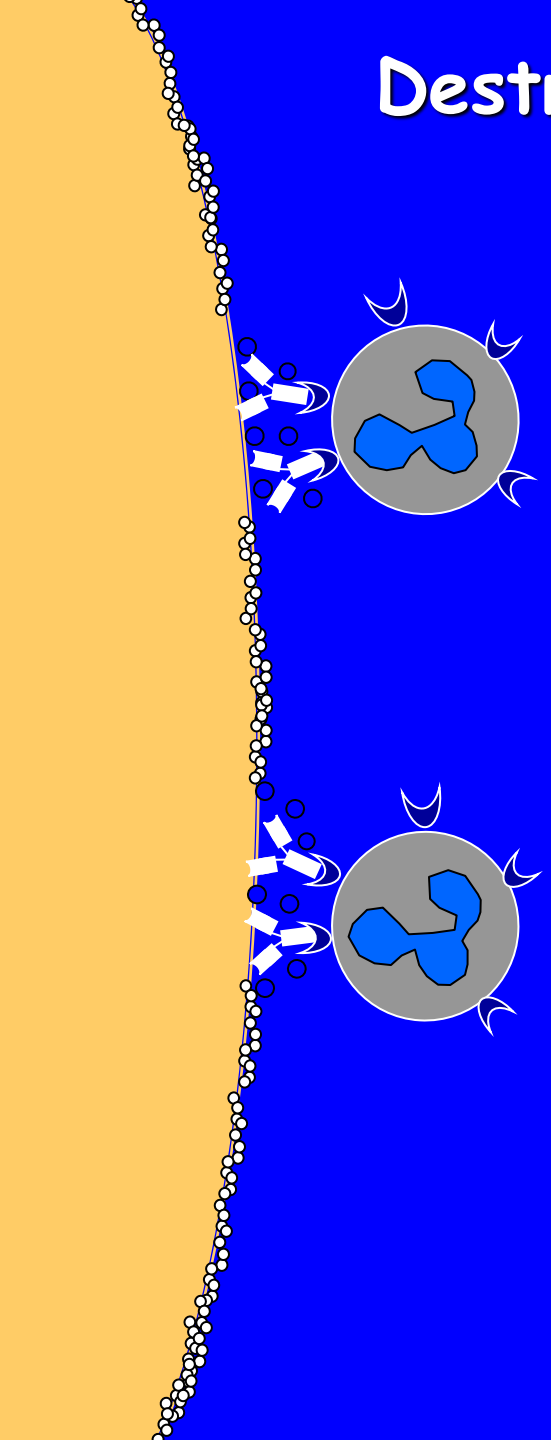
-IL-13, LTs ------ muco

-IL-5 ---------- recrutamento e ativação de eosinófilos

- LTs ---------- contratura da musculatura lisa

Será que o pacote funcional Th2 surgiu para o controle de parasitas (vermes) intestinais?

Destruição de parasitas grandes




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